Lattice Models



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Chapter 1

Prerequisites

1.1 Fourier Analysis

1.1.1 In dimension $d = 1$

For a function $f : \mathbb{Z} \longrightarrow \mathbb{C}$, we define the Fourier series as a 2π-periodic function:

$$\widehat{f}(\xi) = \sum_{k \in \mathbb{Z}} f(k) e^{ik\xi}. \tag{1.1}$$

We can now recall the following theorem:

Theorem 1.1.1. *For two functions $f : \mathbb{Z} \longrightarrow \mathbb{C}$ and $g : \mathbb{Z} \longrightarrow \mathbb{C}$ for which $\widehat{f}$ and $\widehat{g}$ are well-defined (i.e. their corresponding series are absolutely convergent) we have that:*

$$\widehat{f * g}(\xi) = \widehat{f}(\xi)\widehat{g}(\xi). \tag{1.2}$$

Furthermore, for all $n \in \mathbb{Z}$, we have the following Fourier Inversion Formula:

$$f(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \widehat{f}(\xi) e^{-in\xi} d\xi. \tag{1.3}$$

Proof. For the first statement, we manipulate the sums and perform a change of variable:

$$\begin{aligned}
\widehat{f * g}(\xi) = \sum_{k \in \mathbb{Z}} \left(\sum_{l \in \mathbb{Z}} f(l) g(k-l) \right) e^{ik\xi} &= \sum_{k \in \mathbb{Z}} \sum_{l \in \mathbb{Z}} f(l) e^{il\xi} g(k-l) e^{i(k-l)\xi} \\
&= \sum_{l \in \mathbb{Z}} \left(f(l) e^{il\xi} \right) \sum_{k \in \mathbb{Z}} \left(g(k-l) e^{i(k-l)\xi} \right) \\
&= \sum_{l \in \mathbb{Z}} \left(f(l) e^{il\xi} \right) \widehat{g}(\xi) = \widehat{f}(\xi)\widehat{g}(\xi).
\end{aligned}$$

For the second statement, we use the fact that the series are absolutely convergent so that we can exchange summation and integration, i.e. $\sum \int = \int \sum$:

$$\int_{-\pi}^{\pi} \widehat{f}(\xi) e^{-ik\xi} d\xi = \int_{-\pi}^{\pi} \sum_{n\in\mathbb{Z}} f(n) e^{in\xi} e^{-ik\xi} d\xi = \sum_{n\in\mathbb{Z}} f(n) \int_{-\pi}^{\pi} e^{i(n-k)\xi} d\xi$$
$$= \sum_{n\in\mathbb{Z}} f(n)(2\pi\delta(n-k)) = 2\pi f(k).$$

□

1.1.2 In dimension $d \geq 2$

For functions $f : \mathbb{Z}^d \longrightarrow \mathbb{C}$ we have an analogous Fourier series, for any $\underline{\xi} = (\xi_1, ..., \xi_d)$:

$$\widehat{f}(\underline{\xi}) = \sum_{\underline{k}\in\mathbb{Z}^d} f(k) e^{i\underline{k}\cdot\underline{\xi}}. \tag{1.4}$$

Where we still have similar properties as in dimension $d = 1$:

1)
$$(f * g)(\underline{k}) = \sum_{\underline{z}\in\mathbb{Z}^d} f(\underline{z}) g(\underline{k} - \underline{z}). \tag{1.5}$$

2)
$$\widehat{f * g}(\underline{\xi}) = \widehat{f}(\underline{\xi})\widehat{g}(\underline{\xi}). \tag{1.6}$$

3)
$$f(\underline{k}) = \frac{1}{(2\pi)^d} \int ... \int \widehat{f}(\underline{\xi}) e^{-i\underline{k}\cdot\underline{\xi}} d\xi_1 ... d\xi_d. \tag{1.7}$$

We omit the proofs of these three results as they are similar as in the 1-dimensional case presented above.

1.2 Riemann's Mapping Theorem

Let Ω and $\widetilde{\Omega}$ be two Jordan domains (i.e. $\partial\Omega$ and $\partial\widetilde{\Omega}$ are closed simple curves), and let $a, b, c \in \partial\Omega$ and $\widetilde{a}, \widetilde{b}, \widetilde{c} \in \partial\widetilde{\Omega}$ be distinct boundary points, in counterclockwise order.
Then there exists a unique conformal mapping $\phi : \Omega \to \widetilde{\Omega}$ with $a, b, c \to \widetilde{a}, \widetilde{b}, \widetilde{c}$.
In fact, this statement can be generalized to arbitrary simply-connected domains, provided the boundary points are replaced by prime ends.

1.3 Stirling's Approximation

The approximation of $n!$ is given by the following formula:

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \mathcal{O}\left(n^{-1}\right)\right). \tag{1.8}$$

Chapter 2

Simple Random Walks (SRW)

Definition 2.0.1. *A **simple random walk** (SRW) on $G = (V, E)$ starting from $v \in V$ is a sequence $\{X_k\}_{k\geq 0}$ for all integer k, where $X_0 = x$ and X_{k+1} is picked uniformly at random over the neighbours of X_k.*

If we start a walk at the origin on $G = \mathbb{Z}^d$, $d \geq 1$, one of the classical questions concerning SRW is to know if we will, at some point, come back to the starting point with probability 1.
Before coming up with an answer to this question, we present a theorem on the *recurrence* in a SRW, which is the scenario of eventually coming back to the origin.

Theorem 2.0.2. *A SRW $\{X_k\}_{k\geq 0}$ starting at $X_0 = v$ is recurrent if and only if*

$$\sum_{k=0}^{\infty} \mathbb{P}(X_k = v) = +\infty. \tag{2.1}$$

Proof. Consider τ_v^n to be the stopping time for the nth visit at v. Then:

$$\mathbb{P}(\tau_v^n < \infty) = \mathbb{P}(\tau_v^n < \infty | \tau_v^{n-1} < \infty)\mathbb{P}(\tau_v^{n-1} < \infty) = \mathbb{P}(\tau_v^1 < \infty)\mathbb{P}(\tau_v^{n-1} < \infty), \tag{2.2}$$

where we have used the Markov property in the last equality. Hence by recurrence:

$$\mathbb{P}(\tau_v^n < \infty) = \mathbb{P}(\tau_v^1 < \infty)^n. \tag{2.3}$$

Consider now N_v to be the number of visits at v. Then we have that:

$$N_v = \sum_{n=1}^{\infty} \mathbb{1}_{\{\tau_v^n < \infty\}}. \tag{2.4}$$

Hence on the one hand we have:

$$\mathbb{E}(N_v) = \sum_{n=1}^{\infty} \mathbb{P}(N_v \geq n) = \sum_{n=1}^{\infty} \mathbb{P}(\tau_v^n < \infty) = \sum_{n=1}^{\infty} \mathbb{P}(\tau_v^1 < \infty)^n = \frac{\mathbb{P}(\tau_v^1 < \infty)}{1 - \mathbb{P}(\tau_v^1 < \infty)}. \tag{2.5}$$

And on the other hand:

$$\mathbb{E}(N_v) = \mathbb{E}\left[\sum_{n=0}^{\infty} \mathbb{1}_{\{X_n=v\}}\right] = \sum_{n=0}^{\infty} \mathbb{P}(X_n = v). \tag{2.6}$$

This allows us to conclude that $\sum_{n=0}^{\infty} \mathbb{P}(X_n = v) = \infty$ if and only if $\mathbb{P}(\tau_v^1 < \infty) = 1$.

□

Theorem 2.0.3. *(Polyá's Theorem, 1902) Starting at the origin of a SRW on $G = \mathbb{Z}^d$, we will come back to the starting point with probability 1 only if $d \leq 2$.*

The rest of this section will consist of proving Polyá's Theorem. To show this, we want to study the event:

$$\pi_d = \mathbb{P}[\exists k \geq 1 | X_k = 0]. \tag{2.7}$$

An interesting question would be to know if $\pi_d < 1$ or $\pi_d = 1$.
For convenience, we set $N_d = |\{k \geq 1 | X_k = 0\}|$ so that $\pi_d = \mathbb{P}(N_d > 0)$, with the objective of studying a simpler quantity: $\mathbb{E}(N_d)$.

Proof. (In dimension $d = 1$) Since we have that $\mathbb{P}(N_d \geq l) = \mathbb{P}(N_d \geq l | N_d \geq l-1)\mathbb{P}(N_d \geq l-1) = \pi_d \times \pi_d^{l-1}$ and iterating this process l times, we can write the expectation of N_d as:

$$\mathbb{E}(N_d) = \sum_{l=1}^{\infty} l\mathbb{P}(N_d = l) = \sum_{l=1}^{\infty} \mathbb{P}(N_d \geq l) = \sum_{l=1}^{\infty} \pi_d^l = \begin{cases} \frac{1}{1-\pi_d} - 1 = \frac{\pi_d}{1-\pi_d} & \text{,if } \pi_d < 1 \\ \infty & \text{,if } \pi_d = 1 \end{cases}.$$

On the other hand:

$$\mathbb{E}(N_d) = \mathbb{E}\left[\sum_{k=1}^{\infty} \mathbb{1}_{\{X_k=0\}}\right] = \sum_{k=1}^{\infty} \mathbb{P}(X_k = 0). \tag{2.8}$$

To ease the calculations, we now define the matrix $Q_k(x, y)$ to be:

$$Q_k(x, y) = \mathbb{P}(X_k = y | X_0 = x). \tag{2.9}$$

Then it is clear that, denoting by $\underline{0}$ the origin:

$$\mathbb{E}(N_d) = \sum_{k=1}^{\infty} Q_k(\underline{0}, \underline{0}). \tag{2.10}$$

We furthermore note that we can find a recursive formula for Q_k:

$$Q_{k+1}(x, y) = Q_{k+1}(0, y - x) = \sum_{z \in \mathbb{Z}^d} Q_k(x, z) Q_1(z, y). \tag{2.11}$$

Leading to the summation definition for matrix multiplication:

$$Q_{k+1} = Q_k Q_1 \Longrightarrow Q_k = Q_1^k. \tag{2.12}$$

With this matrix formulation in mind, we would like to study the quantity $P_k(z) := Q_k(\underline{0}, z)$, i.e. the probability to be at the vertex $z \in \mathbb{Z}^d$ of the SRW given that we begin the walk at the origin $\underline{0}$. Then:

$$P_{k+1}(z) = \sum_y P_k(y)P_1(z-y) = P_k * P_1 \Longrightarrow P_k = P_1^{*k}, \tag{2.13}$$

where the notation x^{*k} means that x is convoluted k times.
It is now clear that $\mathbb{E}(N_d) = \sum_{k\in\mathbb{N}} P_k(\underline{0}) = \sum_{k\in\mathbb{N}} P_1^{*k}(\underline{0})$, and all it remains is to find an explicit formula for $P_1^{*k}(\underline{0})$. We have:

$$P_1^{*k}(\underline{0}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \widehat{P_1^{*k}}(\xi)d\xi = \frac{1}{2\pi}\int_{-\pi}^{\pi} \left(\widehat{P_1}(\xi)\right)^k d\xi, \tag{2.14}$$

where:

$$\widehat{P_1}(\xi) = \sum_{l\in\mathbb{Z}} P_1(l)e^{il\xi} = \sum_{l\in\mathbb{Z}} \frac{1}{2}\mathbb{1}_{\{-1,1\}}(l)e^{il\xi} = \frac{1}{2}\left(e^{i\xi} + e^{-i\xi}\right) = \cos(\xi). \tag{2.15}$$

Putting things together yields: (See Remark (2.0.4) for interchangeability of $\sum\int = \int\sum$)

$$2\pi\mathbb{E}(N_1) = \sum_{k=0}^{\infty}\int_{-\pi}^{\pi}\cos^k(\xi)d\xi = \int_{-\pi}^{\pi}\sum_{k=0}^{\infty}\cos^k(\xi)d\xi = \int_{-\pi}^{\pi}\frac{1}{1-\cos(\xi)}d\xi = +\infty. \tag{2.16}$$

By Theorem (2.0.2) since $\mathbb{E}(N_1) = +\infty$, it implies that the SRW is recurrent and will come back to the origin with probability 1. This concludes the case $d = 1$.

□

Remark 2.0.4. *(Interchangeability of $\sum\int = \int\sum$) In the above proof, we have freely interchanged sums and integrals. We now justify why we are allowed to perform this step.*
Introduce a mass parameter $\lambda < 1$ and look at the quantity:

$$\sum_{k=0}^{\infty} P_1^{*k}(0)\lambda^k = \sum_{k=0}^{\infty}(\cos(\xi)\lambda)^k. \tag{2.17}$$

We now have an absolutely convergence series, hence we can interchange the sum and the integral in this case:

$$\sum_{k=0}^{\infty}\int_{-\pi}^{\pi}(\cos(\xi)\lambda)^k d\xi = \int_{-\pi}^{\pi}\sum_{k=0}^{\infty}(\cos(\xi)\lambda)^k d\xi = \int_{-\pi}^{\pi}\frac{1}{1-\lambda\cos(\xi)}d\xi. \tag{2.18}$$

Now taking a limit $\lambda \longrightarrow 1$, we can use dominated convergence to interchange integral and limit and we obtain:

$$\lim_{\lambda\to 1}\int_{-\pi}^{\pi}\frac{1}{1-\lambda\cos(\xi)}d\xi = \int_{-\pi}^{\pi}\lim_{\lambda\to 1}\frac{1}{1-\lambda\cos(\xi)}d\xi = \int_{-\pi}^{\pi}\frac{1}{1-\cos(\xi)}d\xi = +\infty. \tag{2.19}$$

On the other hand, the sum $\sum_{k=0}^{\infty} P_1^{*k}(0)\lambda^k$ monotonically converges to $\sum_{k=0}^{\infty} P_1^{*k}(0)$ as $\lambda \longrightarrow 1$, which implies that $\sum_{k=0}^{\infty} P_1^{*k}(0) = +\infty$ as required.

We now shift our interest to the cases where $d \geq 2$.

Proof. (In dimension $d \geq 2$) This proof reuses the construction of $\widehat{P_1}(\underline{\xi})$ in the proof in dimension

1, for a given vector $\underline{\xi} = (\xi_1, ..., \xi_d)$. We then have that:

$$\widehat{P_1}(\underline{\xi}) = \frac{1}{d}(\cos(\xi_1) + ... + \cos(\xi_d)). \tag{2.20}$$

We need to estimate the following multi-dimensional integral:

$$\int_{-\pi}^{\pi} ... \int_{-\pi}^{\pi} \left[\frac{1}{1 - \frac{\cos(\xi_1)+...+\cos(\xi_d)}{d}} \right] d\xi_1 ... d\xi_d = \int_{-\pi}^{\pi} ... \int_{-\pi}^{\pi} \left[\frac{1}{\frac{\xi_1^2+...+\xi_d^2}{d} + o(\|\xi\|^2)} \right] d\xi_1 ... d\xi_d, \tag{2.21}$$

where the second equality comes from a Taylor expansion of $\cos(\underline{\xi})$ around $\underline{0}$.
For $d = 2$:

$$\int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \frac{2}{\xi_1^2 + \xi_2^2} d\xi_1 d\xi_2 = \int_0^{\epsilon} \int_0^{2\pi} \frac{2r}{r^2} dr d\theta = \int_0^{\epsilon} \int_0^{2\pi} \frac{2}{r} dr d\theta = +\infty. \tag{2.22}$$

Hence for $d = 2$, the SRW is recurrent and will come back to the origin with probability 1.

For $d = 3$:

$$\int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \frac{3}{\xi_1^2 + \xi_2^2 + \xi_3^2} d\xi_1 d\xi_2 d\xi_3 = \int_0^{\epsilon} \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 3\cos(\phi) dr d\theta d\phi < +\infty. \tag{2.23}$$

Hence for $d = 3$, the SRW is transient, i.e. it will not come back to the origin with probability 1. Now, if the 4-dimensional SRW returns to the origin with probability 1, a 3-dimensional SRW would also. This is absurd since we just proved that it would not. Iterating this argument to any d-dimensional SRW, we conclude that the walk is transient for any $d > 2$.

□

Proposition 2.0.5. *A simple random walk on G is recurrent when started from v if and only if it is recurrent when started from w.*

Proof. Since G is connected (and locally finite), the probability that $(S_n)_{n\geq 0}$ goes to w before going back to v is strictly positive. Suppose E_k is the event that our walk started from v passes through w during the k-th excursion from v before coming back the $(k+1)$-th time. By the Markov property, the E_k's are independent and have the same non-zero probability. Therefore, an infinite number of E_k happens: with probability 1 the walk from v passes through w infinitely many times. Let us consider the random walk after it goes through w for the first time. Because of the previous discussion, it visits w an infinite number of times, but it has also (by the Markov property) the same law as the random walk which starts at w. So we conclude that the simple random walk on G is recurrent when started from w.

□

Proposition 2.0.6. *A simple random walk on a finite graph is recurrent.*

Proof. We can restrict ourself to the case where G is finite and connected. The walk must be somewhere in G at any time n thus $\sum_{v\in G} \mathbb{1}_{\{S_n=v\}} = 1$. Therefore:

$$\sum_{n\in\mathbb{N}}\sum_{v\in G}\mathbb{1}_{\{S_n=v\}}=\infty. \tag{2.24}$$

Yet,

$$\sum_{n\in\mathbb{N}}\sum_{v\in G}\mathbb{1}_{\{S_n=v\}}=\sum_{v\in G}\sum_{n\in\mathbb{N}}\mathbb{1}_{\{S_n=v\}}=\sum_{v\in G}N_v=\infty, \tag{2.25}$$

where N_v is the number of visits to v. Since G is finite, there exists w such that $N_w=\infty$ with a positive probability. If we consider the random walk conditioned to visit w (which is possible since we know that $N_w=\infty$ and therefore $\mathbb{P}(\exists n, S_n=w)>0$), the law of the walk after the first visit to w is the simple random walk which starts at w. It visits w infinitely many times as the simple random walk is recurrent when started from w and because of Proposition (2.0.5) the simple random walk is recurrent when started from any vertex of G.

□

Chapter 3

Simple Random Walks & PDEs

In this chapter, we aim to explain the connection between simple random walks and the Laplace equation $\Delta f = 0$, and also with the heat equation $\partial_t f = \Delta f$. This connection will be useful to study the continuous limit of random walks, and more generally to understand the Laplace and heat equations.

3.1 Exit Probability in 1-Dimension

Consider the segment $\{1,...,n-1\} \subset \mathbb{Z}$, with boundary $\{0,n\}$ and $B = \{n\}$:



Consider $H(x) := \mathbb{P}(X_\tau^x = n)$, to be the probability that a SRW from $x \in \{0,...,n\}$ hits n before 0, for $\tau = \inf_k\{X_k^x \in \{0,n\}\}$ is the first $k \geq 0$ such that $X_k^x \in \partial G$.
It is clear that $H(0) = 0$ and $H(n) = 1$. Now since the SRW jumps to either $x+1$ or $x-1$ with equal probability $\frac{1}{2}$, once it is there its chances to hit 0 are $H(x-1)$ and $H(x+1)$ by the Markov property. Hence we have a relation for inner vertices, called the *mean value property*:

$$H(x) = \frac{1}{2}H(x-1) + \frac{1}{2}H(x+1) \tag{3.1}$$

Under the appropriate boundary conditions, it is natural to find a solution of the above equation to be $H(x) = \frac{x}{n}$. This solution is in fact the unique one. If we had two, we could take their difference D, which would still satisfy $D(0) = D(n) = 0$ and the mean value property. However, by the *maximum principle*, $D = 0$ otherwise we would have a strict maximum or a strict minimum on $\{1,...,n-1\}$. But that would contradict the mean value property, as a strict maximum/minimum cannot be equal to the values of the two neighbors. In conclusion, the function $H(x) = \frac{x}{n}$ is the unique solution of equation (3.1).

3.2 Discrete Laplacian & the Laplace Equation

Definition 3.2.1. *For a graph G with boundary ∂G (disjoint from G), and given a function $f : \partial G \longrightarrow \mathbb{R}$, the **discrete Laplacian** is a linear map $\Delta_G^f u : \mathbb{R}^G \longrightarrow \mathbb{R}^G$ with boundary condition f is:*

$$\left(\Delta_G^f u\right)(x) = \frac{1}{n(x)} \sum_{y \sim x} [u(y) - u(x)], \tag{3.2}$$

where $n(x) = |\{y : y \sim x\}|$ for $y \in G \cup \partial G$. Here, $y \sim x$ means that x and y are neighbours, and when $y \in \partial G$, $u(y) = f(y)$. We usually consider $n(x) = 2d$ for $G \subseteq \mathbb{Z}^d$.

Now we note that when $f \equiv 0$, then $\Delta_G^f = \Delta_G^0$ is a linear map where:

$$\ker\{\Delta_G^0\} = \{0\}. \tag{3.3}$$

Hence Δ_G^0 is invertible and for all $f : \partial G \longrightarrow \mathbb{R}$, $\Delta_G^f u \equiv 0$ has a unique solution.

Definition 3.2.2. *For given G, ∂G and $f : \partial G \longrightarrow \mathbb{R}$, the **Laplace equation** is:*

$$\Delta_G^f u = 0. \tag{3.4}$$

The functions that satisfy the Laplace equation are **Harmonic functions**. We end this section by claiming that the solutions to the Laplace equation leads to the expectations of exit values for SRWs stopped upon the exit of G.

3.3 Discrete Harmonic Measure

For $(G, \partial G)$ a graph, $B \subseteq \partial G$ and $\tau = \inf\{n : X_n^x \in \partial G\}$ we define the **Harmonic measure** as:

$$H(x, B) := \mathbb{P}(X_\tau^x \in B). \tag{3.5}$$

Now, for $x \in G$, we have that:

$$\Delta H(x, B) = \frac{1}{n(x)} \sum_{y \sim x} H(y, B) \tag{3.6}$$

Hence for all B, $\Delta_G H(\cdot, B) \equiv 0$ everywhere on G and $H(\cdot, B)\big|_{\partial G} = \mathbb{1}_B$ (this is another definition of the Harmonic Measure).
If we now consider $f : \partial G \longrightarrow \mathbb{R}$ and $u(x) := \mathbb{E}[f(X_\tau^x)]$, we obtain that $\Delta u(x) = 0$ for all $x \in G$ and $u\big|_{\partial G} = f$. We can then deduce that the solution to $\Delta_G^f u \equiv 0$ is always $u(x) = \mathbb{E}[f(X_\tau^x)]$.

3.4 Green's Function & the Solution to the Discrete Poisson Equation

Definition 3.4.1. *For $x, y \in G$, the SRW $(X_k^x)_{k\geq 0}$ from x, the stopping time $\tau = \inf\{n : X_n^x \in \partial G\}$, we define the* ***Green Function*** *as:*

$$\mathcal{G}(x,y) := \mathbb{E}\left[|\{0 \leq k < \tau : X_k^x = y\}|\right] = \mathbb{E}\left[\sum_{k=0}^{\tau-1} \mathbb{1}_{\{X_k^x = y\}}\right]. \tag{3.7}$$

This quantity represents the number of visits of y starting from x before hitting ∂G. This quantity is finite for finite graphs.

Consider a finite graph G and its discrete Laplacian $\Delta := \Delta_G^0$. Using the Markov property we have that:

$$(\Delta\mathcal{G})(\cdot, y)(x) = -\delta_{xy} = \begin{cases} -1, & \text{if } x = y \\ 0, & \text{if } x \neq y \end{cases} \tag{3.8}$$

Then the SRW doesn't visit y at time 0 for $x \neq y$ and for $x = y$, at time 0, the visit of y is lost after jumping away by one step.
This means that if we write the operators in a matrix notation we have that $(\Delta\mathcal{G})_{x,y} = -\delta_{xy}$, therefore $\Delta\mathcal{G} = -\mathbb{1} \Longrightarrow \Delta^{-1} := (\Delta_G^0)^{-1} = -\mathcal{G}$. Hence $-\Delta_G^0$ and $\mathcal{G}$ are $|G| \times |G|$ matrices that are inverse of each other.

Definition 3.4.2. *For $f : G \longrightarrow \mathbb{R}$, the* ***discrete Poisson equation*** *is given by:*

$$\begin{cases} \Delta u = f \\ u|_{\partial G} \equiv 0 \end{cases} \tag{3.9}$$

Proposition 3.4.3. *The solution to the Discrete Poisson Equation is given in terms of the Green's function $\mathcal{G}$ by:*

$$u = \Delta^{-1} f = -\mathcal{G} f. \tag{3.10}$$

Remark 3.4.4. *We have the following probabilistic interpretation of $-\mathcal{G}f$:*

$$u(x) = -\mathcal{G}f = \mathbb{E}\left[\sum_{n=0}^{\tau-1} (-f)(X_n^x)\right]. \tag{3.11}$$

3.5 Discrete Heat Equation

In this section, consider a SRW $(X_k^x)_{k\geq 0}$ and set $u(x,t)$ to be:

$$u(x,t) := \mathbb{P}\left(\forall k \leq t : X_k^x \in G\right). \tag{3.12}$$

Here $u(x,t)$ is the probability that a SRW starting at x stays in G until at least time t. Furthermore, we have that:

$$u(x,t+1) = \frac{1}{4}\sum_{y\sim x} u(y,t), \tag{3.13}$$

and we write:

$$[d_t]u(x,t) := u(x,t+1) - u(x,t). \tag{3.14}$$

Definition 3.5.1. *Consider a finite graph G with boundary ∂G. Let $(X_k^x)_{k\geq 0}$ be a SRW and $\Delta := \Delta_G^0$. The **Discrete Heat Equation** is given by:*

$$[d_t]u(x,t) = \Delta u(x,t), \tag{3.15}$$

with boundary conditions $u(\cdot|_{\partial G},t) = 0$ and $u(\cdot|_G,0) = 1$.

Remark 3.5.2. *The solution to the Discrete Heat Equation is unique.*

3.6 Convergence of PDE Solutions

Take a lattice $\mathbb{Z}^d$, a domain $\Omega \in \mathbb{R}^d$ and look at $\Omega_\delta = \Omega \cap \delta\mathbb{Z}^d$ (e.g. largest connected component). Here $\delta\mathbb{Z}^d$ corresponds to a lattice rescaled by $\delta > 0$ and,

$$\partial\Omega_\delta = \{x \in \delta\mathbb{Z}^d \backslash \Omega_\delta : x \sim \Omega_\delta\}. \tag{3.16}$$

Now, on the convergence of operators, if $f : \Omega \longrightarrow \mathbb{R}$ is C^2 we have,

$$\Delta_\delta f(x) = \frac{1}{2d}\left[\sum_{i=1}^{d}\sum_{s\in\{\pm 1\}} f(x+se_i\delta) - f(x)\right]. \tag{3.17}$$

Taylor expanding $f(x+se_i\delta)$ gives us:

$$f(x+se_i\delta) = f(x) + s\delta\partial_i f(x) + \frac{1}{2}\delta^2 s^2 \partial_{ii} f(x) + o(\delta^2). \tag{3.18}$$

Therefore, as $\delta \longrightarrow 0$ and denoting by $\Delta := \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2}$ we obtain:

$$\frac{1}{\delta^2}\Delta_\delta f(x) \longrightarrow \frac{1}{2d}\Delta f. \tag{3.19}$$

Now the problem at hand is to know if discrete solutions of discrete PDEs converge. The standard method to solve this problem is to study precompactness and uniqueness. We can study precompactness via the Harnack inequalities, after which uniform continuity up to boundary can be studied using Beurling's inequality and uniqueness will follow with a continuous maximum principle.

3.6.1 Discrete Harnack Inequality

Theorem 3.6.1 (Discrete Harnack Inequality). *If we have a discrete domain $\Omega_\delta \subset \delta\mathbb{Z}^d$ and a discrete harmonic function $f : \Omega_\delta \cup \partial\Omega_\delta \longrightarrow \mathbb{R}$, then for all $i = \{1, ..., d\}$ and for some constant C_1,*

$$\left|\frac{f(x+\delta e_i) - f(x)}{\delta}\right| \leq C_1 \frac{\max_{z\in\partial\Omega_\delta} |f(z)|}{\operatorname{dist}(x, \partial\Omega_\delta)}. \tag{3.20}$$

Sketch of Proof. Let us find a bound for the following quantity:

$$(*) = \left|\frac{f(x+\delta e_i) - f(x-\delta e_i)}{2\delta}\right| = \frac{1}{2\delta}\left|\mathbb{E}\left[f\left(X_\tau^{x+\delta e_i}\right) - f\left(\widetilde{X}_{\widetilde{\tau}}^{x-\delta e_i}\right)\right]\right|, \tag{3.21}$$

where we take $\widetilde{X}$ to be the mirror of X until the axis is hit and equal to X after. Now:

$$(*) \leq \frac{1}{2\delta}\mathbb{P}\left(X_\tau^{x+\delta e_i} \text{ did not hit the mirror}\right)\left(2 \max_{z\in\partial\Omega_\delta} |f(z)|\right) \tag{3.22}$$

The last step is to bound above the probability $\mathbb{P}^*$ to not hit the mirror. Then, for some constant C_2, one can show that:

$$\mathbb{P}^* \leq \mathbb{P}\left(\text{Hit circle before the mirror}\right) \leq C_2 \frac{\delta}{\operatorname{dist}(z, \partial\Omega_\delta)} \tag{3.23}$$

Combining equations (3.22) and (3.23) we obtain the required inequality.

□

3.6.2 Discrete Beurling Inequality in 2-Dimensions

Theorem 3.6.2 (Discrete Beurling Inequality). *Let Ω be a 2-dimensional bounded domain, Ω_δ its discretisation and $b_\delta \subset \partial\Omega_\delta$ the corresponding boundary. Then for the Harmonic measure $H(b_\delta, z)$ we have:*

$$H(b_\delta, z) \leq C_3 \left(\frac{\operatorname{dist}(z, \partial\Omega_\delta \backslash b_\delta)}{\operatorname{dist}(z, b_\delta)}\right)^\alpha, \tag{3.24}$$

for some constants $C_3, \alpha > 0$ and for all δ, Ω.

Sketch of Proof. Assume z is closer to $\partial\Omega_\delta \backslash b_\delta := c_\delta$ than to b_δ. Construct n concentric annuli of radii $r = \operatorname{dist}(z, c_\delta)$ where $3r, 9r, 27r, ..., 3^{k+1}r < \operatorname{dist}(z, b_\delta)$. We can construct k such annuli with, $k = \lfloor \log_3(R/r) \rfloor$, where $R = \operatorname{dist}(z, b_\delta)$. In each annulus there is a probability to make a loop upon entering and before leaving it that is greater than some uniform $\epsilon > 0$. But then we would hit c_δ. Now with a couple of manipulations we obtain:

$$\mathbb{P}\left(\text{Hit } b_\delta \text{ before } c_\delta\right) \leq (1-\epsilon)^k \leq \text{Constant}\left(\frac{r}{R}\right)^\alpha. \tag{3.25}$$

□

Chapter 4

Uniform Spanning Trees (UST)

In this chapter we will study uniform spanning trees via the use of loop erased random walks and the Wilson's algorithm.

Definition 4.0.1. *For a graph $G=(V,E)$, a **spanning tree** is a subgraph $\widetilde{G}=(\widetilde{V},\widetilde{E})$ that is a tree and contains all vertices of G. Then, given a graph G, a **uniform spanning tree** (UST) is a spanning tree picked out at random.*

For $G=(V,E)$ a graph, label vertices $v_1,...,v_n$ in an arbitrary order. We first define the notion of Loop Erased Random Walk (LERW).

Definition 4.0.2. *A **Loop Erased Random Walk (LERW)** from x to $S\subset V\backslash\{x\}$ is a SRW from x on G, stopped upon hitting S and erase the cycles formed by the loop in a chronological order.*

Proposition 4.0.3 (Wilson's Algorithm)**.** *Wilson's Algorithm constructs USTs by defining the random sequence of trees $\mathcal{T}$ as follows:*
1) $\mathcal{T}_1=\{v_1\}$
2) $\mathcal{T}_2=\mathcal{T}_1\cup\text{LERW}(v_2,\mathcal{T}_1)$
3) $\mathcal{T}_3=\mathcal{T}_2\cup\text{LERW}(\text{First vertex in } V\backslash\mathcal{T}_2,\mathcal{T}_2)$
...
k) Until $\mathcal{T}_k$ contains all vertices of V.

Lemma 4.0.4. *If a cycle C is erasable and we instead choose to erase cycles C_1 and C_2 that are both different from C then C is still erasable.*

Proof. If the cycles $C_1,...,C_l$ are (vertex-) disjoint from C then C is still erasable. Let us argue by contradiction.
Assume this was not true, then there would be a first k such that $C_k\cap C\neq\varnothing$. Now pick $v\in C_k\cap C$ and follow the arrows. Since none of the arrows of C have been erased, following arrows recovers C. Hence $C=C_k$, which contradicts the assumption.

□

Theorem 4.0.5. *Wilson's Algorithm gives a spanning tree with uniform probability.*

Proof. Consider on each vertex $v \in V \backslash \{v_1\}$ an infinite stack of arrows with each arrow of v pointing to a random neighbour of v, uniformly and independently. If we look at the arrows at the top of the stack, we have a tree containing v_1 (with arrows pointing towards v_1) and cycles. For each cycle we may erase its arrows, i.e. get rid of the top of the stacks of its vertices and replace them by the arrows below (in the stacks).
Now, by Lemma (4.0.4), the order in which we erase the cycles at the top of the stacks is irrelevant. If at some point we have nothing to erase then we have a tree.

□

Remark 4.0.6. *We cannot have two cycles touching each other (i.e. issued from the same vertex) since each vertex can only choose a single direction.*

Spanning trees of G are in one-to-one correspondence with arrow-defined trees (with special vertex v_1). Furthermore, the cycle erasing algorithm terminates with probability 1 since the space of arrow configurations at the top of the stack is finite.
Each configuration $\{\text{Cycles } C_1, ..., C_l \text{ erased} + \text{Tree } \mathcal{T}\}$ has the same probability as another configuration $\{\text{Cycles } C_1, ..., C_l \text{ erased} + \text{Tree } \widetilde{\mathcal{T}}\}$ since each arrow is i.i.d. uniform. Looking at this statement in an other way we observe that:

$$\mathbb{P}(\mathcal{T}) = \sum_{\text{Cycles } C_1,...,C_l \text{ erased}} \mathbb{P}\{\text{Cycles } C_1, ..., C_l \text{ erased} + \text{Tree } \mathcal{T}\}. \tag{4.1}$$

Hence we have that, for all trees $\mathcal{T}, \widetilde{\mathcal{T}}$, $\mathbb{P}(\mathcal{T}) = \mathbb{P}(\widetilde{\mathcal{T}})$, which implies that $\mathcal{T}$ is uniform. Concluding, Wilson's Algorithm is equivalent to exploring vertices following arrows and erasing cycles we find along the way.

Proposition 4.0.7 (Euler's Formula)**.** *For V, E, F the number of vertices, edges, and faces of G, we have that $V - E + F = 2$.*

Proof. Recall that for a planar graph we define its dual $G' = (V', E')$ that has a vertex for each face of G and has an edge for each pair of faces in G that are separated from each other by an edge, and a self-loop when the same face appears on both sides of an edge. Thus, each edge e of G has a corresponding dual edge, whose endpoints are the dual vertices corresponding to the faces on either side of e.
(1) Now, let T be a spanning tree of G and let us consider its edge complement $T^c = E \backslash T$. The edges in the dual graph corresponding to T^c also form a spanning tree. Indeed, let us denote by T' the subgraph in G' induced by T^c. Notice that if T contained a loop, this would isolate a set of faces in T'. So since T does not contain any loop, each vertex in the dual graph is in one connected component of T'. Similarly, if T' contained a loop, the loop would isolate a set of vertices of G and T could not be spanning the whole graph. Hence, T' is loop-less and thus, a spanning tree in the dual graph.

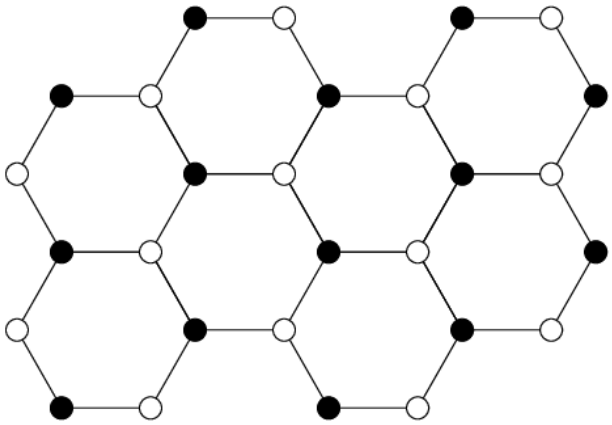
(2) Recall also that each tree has a leaf. Hence, by induction on the size of the tree, one obtains that a tree with k vertices has $k-1$ edges.
Now for the proof of the proposition, we pick a spanning tree T of G and by $E(T)$ we denote the number of edges of T. Then, $E = E(T) + E(T^c)$. Using (2) we know that $V = E(T) + 1$. Using (1), we know that T^c is a spanning tree of the dual graph, and thus $F = E(T^c) + 1$. Together, this gives that $E = V + F - 2$.

□

Chapter 5

Percolation & Cardy's Formula

Percolation is originally a model of a porous medium. We can think that black hexagons are filled with matter, while white ones are empty. We consider the Honeycomb lattice where we randomly color the faces in black or in white with probability p and $1-p$ independently of others.



One of the questions that we can consider is to know if we can go from a vertex A to a vertex B using only black hexagons $A \longleftrightarrow_b B$ or with white hexagons $A \longleftrightarrow_w B$.

Theorem 5.0.1 (Kesten, 1982)**.** *$p > 1/2 \iff$ There exists an infinite connected component of black hexagons (and if it exists, it is unique).*

Theorem 5.0.2 (Smirnov, 2001)**.** *Let $p = \frac{1}{2}$ and take a Jordan domain of $\mathbb{C}$ (domain whose boundary is a closed curve), discretize it by hexagons of size $\delta > 0$ in Ω_δ and fix four points on $\partial\Omega$. Consider the function $F_\delta(\Omega, a, b, c, d) := \mathbb{P}([a,b] \longleftrightarrow_b [c,d])$.*
Then F_δ has a limit as $\delta \to 0$ which is conformally invariant; i.e. if $\phi : (\Omega, a, b, c, d) \to (\widetilde{\Omega}, \widetilde{a}, \widetilde{b}, \widetilde{c}, \widetilde{d})$ is conformal (bijective and holomorphic function) we have:

$$\lim_{\delta \to 0} F_\delta(\Omega, a, b, c, d) = \lim_{\delta \to 0} F_\delta(\widetilde{\Omega}, \widetilde{a}, \widetilde{b}, \widetilde{c}, \widetilde{d}). \tag{5.1}$$

As a result of Smirnov's Theorem, in this section we present and prove Cardy's Formula.

Theorem 5.0.3 (Cardy's Formula)**.** *As a result of Smirnov's Theorem, if Ω is an equilateral triangle with vertices $a=1, b=i\frac{\sqrt{3}}{3}, c=-i\frac{\sqrt{3}}{3}$, then:*

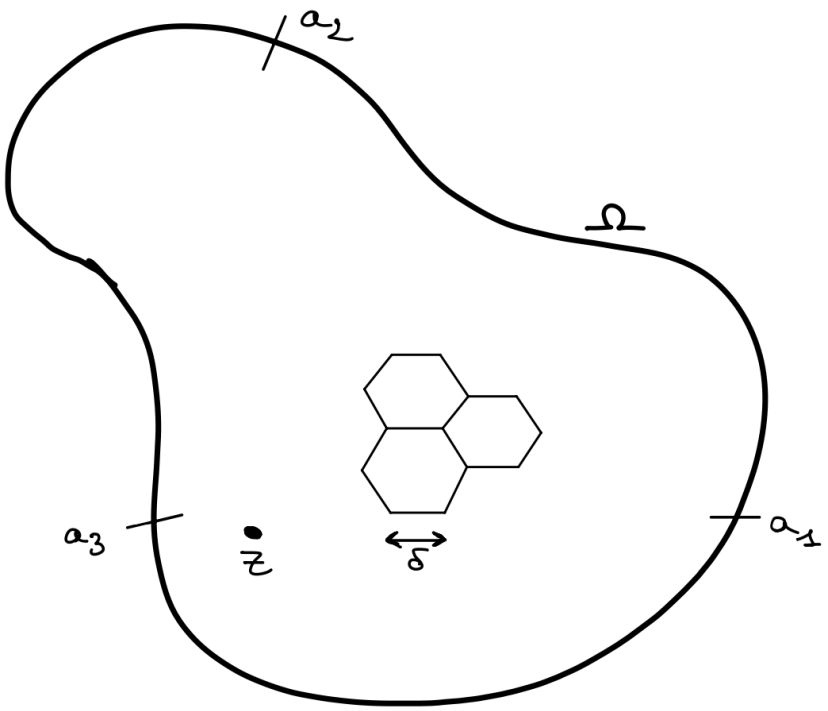
$$\lim_{\delta\to 0} F_\delta(\Omega,a,b,c,d)=\frac{|c-d|}{|c-a|}. \tag{5.2}$$

To prove Cardy's Formula, we first aim to prove the following statement. The formula will then hold by evaluating the proposition at $a_4\in[a_3,a_1]$.

Proposition 5.0.4. *Let (Ω,a_1,a_2,a_3) be a Jordan domain with a_1,a_2,a_3 in trigonometric order. For $z\in\overline{\Omega}$, we first define the two following sets $Q_i^\delta(z)=\{a_i$ and z are separated from a_{i+1} and a_{i+2} by a black path$\}$ for $i=\{1,2,3\}$ (indices modulo 3), and $H_i^\delta(z):=\mathbb{P}\Big(Q_i^\delta(z)\Big)$. Then:*

$$\phi_\delta:=H_1^\delta(z)+\frac{i\sqrt{3}}{3}H_2^\delta-\frac{i\sqrt{3}}{3}H_3^\delta(z)\xrightarrow{\delta\to 0}H(z), \tag{5.3}$$

uniformly on compact sets of $\overline{\Omega}$ and where H is the unique conformal mapping $\Omega\to\triangleright$, for $\triangleright$ the equilateral triangle with vertices $1,\frac{i\sqrt{3}}{3},-\frac{i\sqrt{3}}{3}$ which are the images of a_1,a_2,a_3.



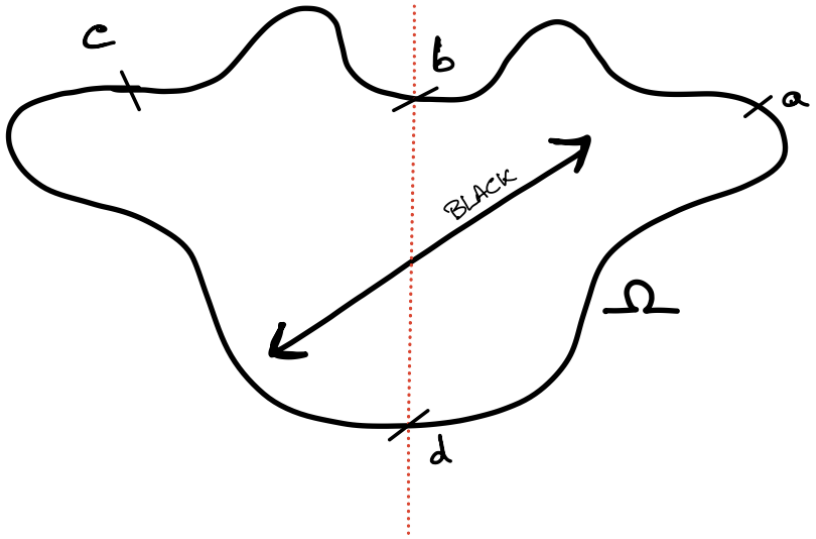
The proof of the above Proposition can be done in three steps. We first have to show precompactness, where in particular we will demonstrate that $(\phi_\delta)_{\delta>0}$ is equicontinuous on $\overline{\Omega}$. Letting ϕ be a subsequential scaling limit $\phi_{\delta_n}\to\phi$, uniqueness of the limit will yield the full convergence. Next, we turn to boundary conditions where we have to show that if $\phi_{\delta_n}\to\phi$, then ϕ is a homeomorphism $\partial\Omega\to\partial\triangleright$ with $a_1,a_2,a_3\mapsto 1,\frac{i\sqrt{3}}{3},-\frac{i\sqrt{3}}{3}$. Finally we will consider analyticity where we will show that if $\phi_{\delta_n}\to\phi$, then ϕ is analytic. This will follow from approximate discrete Cauchy-Riemann relations. To conclude, i.e. showing that $\phi:\Omega\to\triangleright$ is a well-defined bijection inside Ω, we will use the argument principle where we will take $w\in\mathbb{C}$ and show that $\phi(z)-w$ has a single zero in Ω if and only if $w\in\triangleright$.

5.0.1 Russo-Seymour-Welsh Estimates (RSW)

Similarly for the random walks, we first need some simple a priori estimates for the precompactness part.

Lemma 5.0.5. *For a domain Ω with an axis of symmetry through b and d:*

$$\mathbb{P}_\delta([a,b] \longleftrightarrow_b [c,d]) \xrightarrow{\delta\to 0} \frac{1}{2}. \tag{5.4}$$



Proof. (See Bollobás-Riordan, 2006)
The complementary of this event is precisely $[b,c] \longleftrightarrow_w [d,a]$ (a topological fact), and both events have the same probability as $\delta \to 0$. □

To prove Cardy's formula we need to introduce multiple results that will serve as ingredients for the proof.

Theorem 5.0.6 (Russo-Seymour-Welsh Estimates (RSW))**.** *For a (2×1) rectangle:*

$$\mathbb{P}\left(\alpha \xleftarrow{\text{BLACK}} \beta\right) = \mathbb{P}(\textit{Linking edges } \alpha \textit{ to } \beta \textit{ by a black path}) \geq \frac{1}{16}. \tag{5.5}$$

Theorem 5.0.7 (FKG Inequality)**.** *If A and B are two black-increasing events (i.e. If we have the event occur in a configuration and we turn some white hexagons black, they still occur) then:*

$$\mathbb{P}(A\cap B) \geq \mathbb{P}(A)\mathbb{P}(B) \tag{5.6}$$

(or $\mathbb{P}(A|B) \geq \mathbb{P}(A)$).

To prove FKG, we need the following Lemma.

Lemma 5.0.8. *If f,g are two increasing functions $\mathbb{R} \to \mathbb{R}$ and μ is a probability measure:*

$$\int f g d\mu \geq \left(\int f d\mu\right)\left(\int g d\mu\right). \tag{5.7}$$

Proof. On the one hand:

$$\int\int (f(x)-f(y))(g(x)-g(y))d\mu(x)d\mu(y) \geq 0. \tag{5.8}$$

And on the other hand:

$$\int\int (f(x)-f(y))(g(x)-g(y))d\mu(x)d\mu(y) = \int f g d\mu + \int f g d\mu - \left(\int f d\mu\right)\left(\int g d\mu\right) - \left(\int f d\mu\right)\left(\int g d\mu\right).$$

Compiling the two results yields:

$$\int f g d\mu \geq \left(\int f d\mu\right)\left(\int g d\mu\right). \tag{5.9}$$

□

Remark 5.0.9. *This lemma also works if f,g are functions $\mathbb{R}^n \to \mathbb{R}$ non-decreasing with respect to each coordinate and μ is a product of probability measures.*

Proof. (FKG)
To prove FKG we represent the color black as 1, the color white as 0 and the indicator functions $\mathbb{1}_A$,$\mathbb{1}_B$ as non-decreasing functions with respect to each coordinate. Then:

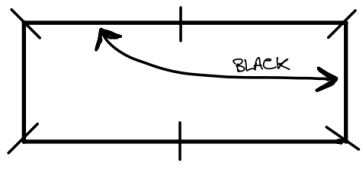
$$\mathbb{E}[\mathbb{1}_A \mathbb{1}_B] = \mathbb{P}(A \cap B) \geq \mathbb{P}(A)\mathbb{P}(B) = \mathbb{E}[\mathbb{1}_A]\mathbb{E}[\mathbb{1}_B]. \tag{5.10}$$

□

We now have all the ingredients to prove the RSW Estimates.

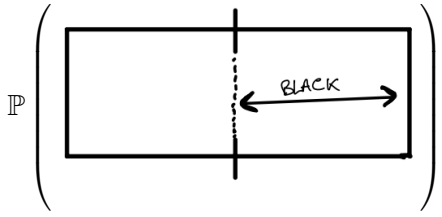
Proof. (RSW) • We first prove that:

$$\mathbb{P}\left(\right) \geq \frac{1}{4}. \tag{5.11}$$



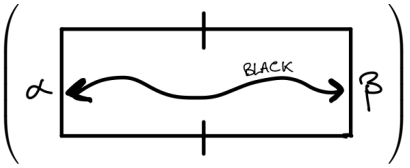
We can observe that the following equality holds:

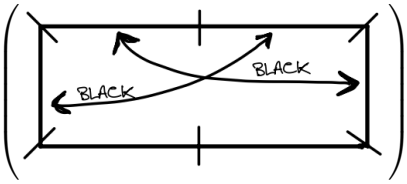
$$\mathbb{P}\left(\quad\right) = \frac{1}{2}. \tag{5.12}$$



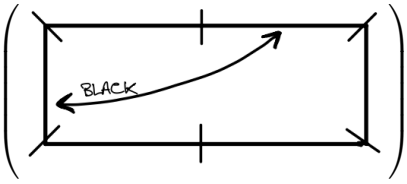
Conditionally on this event, take the lowest black path linking the middle vertex side to the right vertex side. Everything above that is independent. Taking the reflection of that path with respect to the middle vertex side and considering the part of the rectangle above it, there is a probability of $\frac{1}{2}$ of having a black path from top-left to either the first black path or the right vertex. In these cases, with probability $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ we found a black path, as desired.
From there we apply FKG to find:

$$\mathbb{P}\left(\quad\right) \geq \mathbb{P}\left(\quad\right)$$

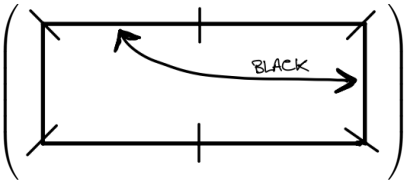




$$\geq \mathbb{P}\left(\quad\right)$$



$$\times\, \mathbb{P}\left(\quad\right)$$



$$\geq \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

• We can also give a more mathematical formalism for the steps described above. Consider a rectangle $R = [0,2] \times [0,i]$ and a symmetric discretization R_δ. We aim to show that:

$$\mathbb{P}_{R_\delta}\left([0,i] \longleftrightarrow_b [2,2+i]\right) \geq \frac{1}{16}. \tag{5.13}$$

With probability $\frac{1}{2}$ there is a path $\gamma : [0,1] \longleftrightarrow [1,1+i]$ made only of black hexagons, and take the lowest such path possible. Suppose that everything above it is independent percolation.
Now, let $\widetilde{\gamma}$ be the reflection of γ with respect to the line $1+i\mathbb{R}$ and let D_δ be the connected component $R_\delta \backslash (\gamma \cup \widetilde{\gamma})$ lying above $\gamma \cup \widetilde{\gamma}$ and intersecting the line $1+i\mathbb{R}$. With probability $\frac{1}{2}$ there is a path $\lambda \subset D_\delta$ linking the bottom-left part of ∂D_δ to the top right one, again by symmetry. Joining γ and λ yields a black path $[0,i] \longleftrightarrow_b [1+i,2+i]$. Therefore, $\mathbb{P}_{R_\delta}([0,i] \longleftrightarrow_b [1+i,2+i]) \geq \frac{1}{4}$, and by symmetry we also have $\mathbb{P}_{R_\delta}([i,1+i] \longleftrightarrow_b [2,2+i]) \geq \frac{1}{4}$. If both of these events occur, then there is a black path $[0,i] \longleftrightarrow_b [2,2+i]$, and by FKG their intersection is greater than $\frac{1}{16}$.

□

5.0.2 Precompactness

We first start with a result on the Hölder continuity of H_i^δ.

Proposition 5.0.10. *Each H_i^δ is uniformly Hölder with respect to the induced domain metric $d_\Omega(x,y) = \inf_{\gamma: x\to y \text{ in } \overline{\Omega}} \text{length}(\gamma)$, i.e.*

$$\left|H_i^\delta(z) - H_i^\delta(w)\right| \leq C\left(d_\Omega(z,w)\right)^\alpha. \tag{5.14}$$

Proof. For $i=1$ we have that:

$$\begin{aligned}\left|\mathbb{P}\left(Q_1^\delta(z)\right) - \mathbb{P}\left(Q_1^\delta(w)\right)\right| &= \left|\mathbb{P}\left(Q_1^\delta(z)\backslash Q_1^\delta(w)\right) - \mathbb{P}\left(Q_1^\delta(w)\backslash Q_1^\delta(z)\right)\right| \\ &\leq \mathbb{P}\left(Q_1^\delta(z)\backslash Q_1^\delta(w)\right) + \mathbb{P}\left(Q_1^\delta(w)\backslash Q_1^\delta(z)\right).\end{aligned}$$

Now, if $Q_1^\delta(z)\backslash Q_1^\delta(w)$ occurs then this implies that a (small) circle of diameter $d_\Omega(z,w)$ is connected by a monochromatic path to each of $[a_1,a_2],[a_2,a_3]$ and $[a_3,a_1]$. Now it holds that:

$$\inf_{z\in\overline{\Omega}}\left(\max_{i=1,2,3}\text{dist}(z,[a_i,a_{i+1}])\right) = \eta > 0. \tag{5.15}$$

Then the probability of at least one of these monochromatic connections is upper bounded by $C\left(\frac{d_\Omega(z,w)}{\eta}\right)^\alpha$ since from RSW, $\mathbb{P}(\text{Monochromatic Connection}) \leq C\left(\frac{r}{R}\right)^\alpha$, for $C,\alpha>0$.
Therefore, $\mathbb{P}\left(Q_1^\delta(z)\backslash Q_1^\delta(w)\right)$ and $\mathbb{P}\left(Q_1^\delta(w)\backslash Q_1^\delta(z)\right)$ are both relatively small, and this happens only if $d_\Omega(z,w)$ is also small.

□

To extract converging subsequences, we extend ϕ_δ into a continuous function (e.g. by piecewise affine interpolation), and use Arzelà-Ascoli Theorem to have that $(\phi_\delta)_\delta$ is bounded and equicontinuous.
From now on, we will assume that ϕ is a subsequential scaling limit $\lim_{\delta_n\to 0}\phi_{\delta_n}$, with $\phi = H_1 + i\frac{\sqrt{3}}{3}H_2 - i\frac{\sqrt{3}}{3}H_3$.

5.0.3 Boundary Conditions

We now want to prove that ϕ is a homeomorphism $\partial\Omega \to \partial\triangleright$, hence wanting to show that it is a homeomorphism $A_1 \to \left[\frac{i\sqrt{3}}{3}, -\frac{i\sqrt{3}}{3}\right]$, $A_2 \to \left[\frac{-i\sqrt{3}}{3}, 1\right]$ and $A_3 \to \left[1, \frac{i\sqrt{3}}{3}\right]$.
For $Q_1^\delta(z)$, if $z\in[a_2,a_3]$, it implies that z is connected to both $[a_1,a_2]$ and $[a_1,a_3]$ by a black path, and at least one of the two is at distance $\geq d>0$. So by RSW this probability converges to 0. Therefore, it suffices to prove that $[a_2,a_3]$ gets mapped to $\left[\frac{i\sqrt{3}}{3}, -\frac{i\sqrt{3}}{3}\right]$. We can show this fact in three propositions:

Proposition 5.0.11. *For any $z\in A_1$:*

$$H_1^\delta(z) = 0. \tag{5.16}$$

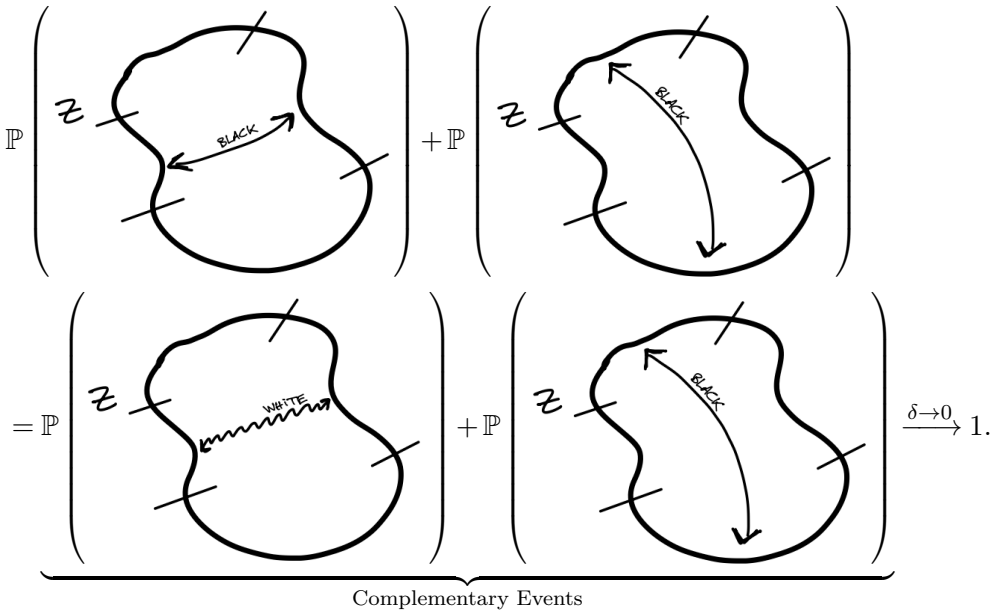
Proof. This result follows from RSW. If $z\in\Omega_\delta$ is at microscopic distance from A_1, then if $Q_1^\delta(z)$ happens, there is a black path from z to A_2 and to A_3. At least one of A_2 and A_3 is at macroscopic

distance from z. Hence $\mathbb{P}_{\Omega_\delta}\left(Q_1^\delta(z)\right) \to 0$. □

Proposition 5.0.12. *For any $z \in A_1$:*

$$H_2^\delta(z) + H_3^\delta(z) \xrightarrow{\delta \to 0} 1. \tag{5.17}$$

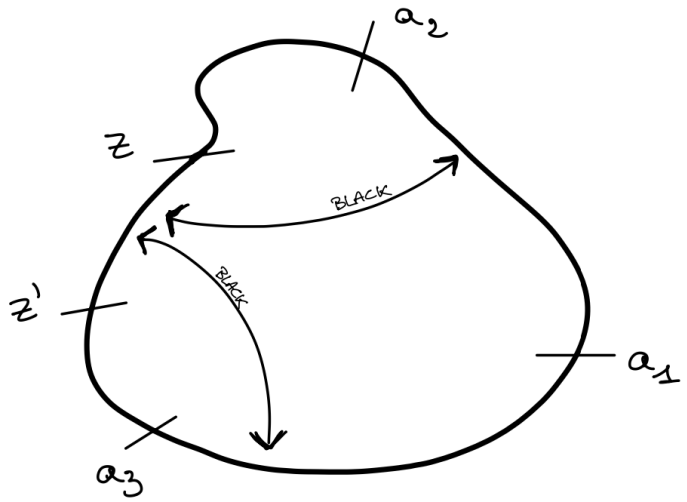
Proof. We have that:



$\xrightarrow{\delta \to 0} 1.$

□

Proposition 5.0.13. *As $z \in A_1$ moves from a_2 to a_3, $H_3^\delta(z)$ increases from 0 to 1.*



Proof. Let $z, z' \in A_1$ with z closer to a_2 than z'. We have $Q_3^\delta(z) \subset Q_3^\delta(z')$, so let $R_\delta := Q_3^\delta(z') \backslash Q_3^\delta(z) = \{[z, z'] \longleftrightarrow_b A_2\}$. By RSW, the probability of the latter event is strictly positive, so $H_3^\delta(z') -$

$H_3^\delta(z) = Q_3^\delta(z')\backslash Q_3^\delta(z)$ is uniformly positive as $\delta \to 0$. Therefore, by RSW and making concentric annuli, we have that $H_3^\delta(a_3) \to 1$ as $\delta \to 0$ (we can make concentric annuli around a_3). □

5.0.4 Discrete Cauchy-Riemann Equations

The final step will be to prove Analyticity. Before we do so, we need to introduce the notions of discrete derivatives and the discrete Cauchy-Riemann equations.

Definition 5.0.14. *For an oriented edge $\overrightarrow{e} \in \Omega_\delta$ from vertex $x \in \Omega_\delta$ to vertex $y \in \Omega_\delta$ and a function $f : \Omega_\delta \longrightarrow \mathbb{C}$ we define the* ***discrete derivative*** $\partial_{\overrightarrow{e}} f$ *by* $f(y) - f(x)$.

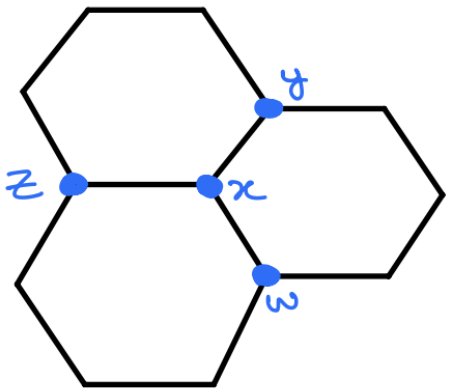
For the function H_μ^δ, we can write:

$$\partial_{\overrightarrow{e}} H_\mu^\delta = \partial_{\overrightarrow{e}}^+ H_\mu^\delta - \partial_{\overrightarrow{e}}^- H_\mu^\delta, \tag{5.18}$$

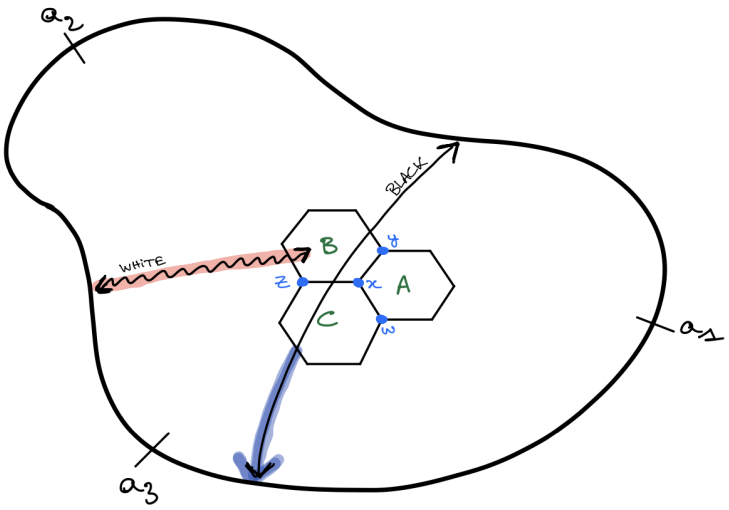
where $\partial_{\overrightarrow{e}}^+ H_\mu^\delta = \mathbb{P}\left(Q_\mu^\delta(y)\backslash Q_\mu^\delta(x)\right)$ and $\partial_{\overrightarrow{e}}^- H_\mu^\delta = \mathbb{P}\left(Q_\mu^\delta(x)\backslash Q_\mu^\delta(y)\right)$. Furthermore, set $\tau = e^{2\pi i/3}$ and write $\tau\overrightarrow{e}$ for the rotation of $\overrightarrow{e}$ around its origin x by $2\pi/3$, $\tau^2 = e^{4\pi i/3}$ to write $\tau^2\overrightarrow{e}$ for the rotation of $\overrightarrow{e}$ around its origin x by $4\pi/3$, and finally we set $H_1^\delta, H_2^\tau, H_3^{\tau^2} := H_1^\delta, H_2^\delta, H_3^\delta$. We are now ready to present the discrete Cauchy-Riemann equations.

Proposition 5.0.15 (The Discrete Cauchy-Riemann Equations)**.** *For $\mu \in \{1, \tau, \tau^2\}$ and an oriented edge $\overrightarrow{e}$, we have:*

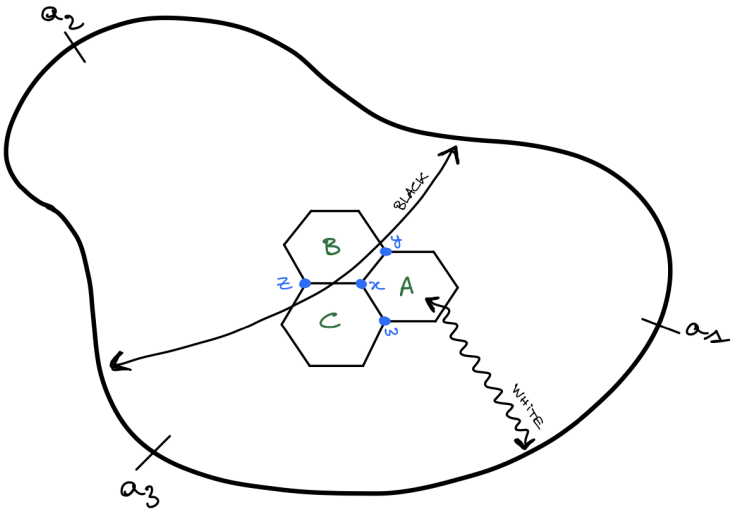
$$\partial_{\overrightarrow{e}}^+ H_\mu^\delta = \partial_{\tau\overrightarrow{e}}^+ H_{\tau\mu}^\delta = \partial_{\tau^2\overrightarrow{e}}^+ H_{\tau^2\mu}^\delta. \tag{5.19}$$



Proof. Since the identity is symmetric, we can simply prove one equality.



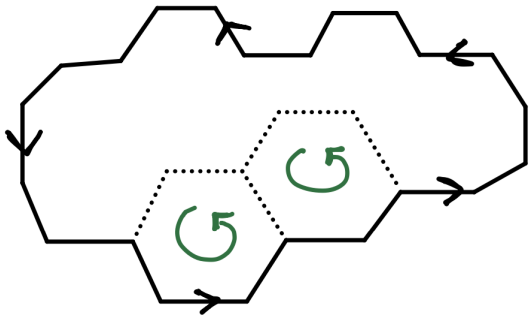
We can then show that $\partial^+_{\overrightarrow{e}} H_1^\delta = \partial^+_{\tau\overrightarrow{e}} H_2^\delta$. The idea is to construct a bijection from $Q_1^\delta(y)\backslash Q_1^\delta(x)$ to $Q_1^\delta(x)\backslash Q_1^\delta(y)$. Each of the white paths has a probability of $\left(\frac{1}{2}\right)^{\#\text{ hexagons}}$.To construct a bijection we do the following: Take the clockwise most black path from C to $[a_3, a_1]$, the counterclockwise most white path from B to $[a_2, a_3]$. Flip the colors, black $\longleftrightarrow$ white, in the region containing $[a_1, a_2]$ bordered by these two paths themselves. Finally flip all the colors again.
This yields a bijection as it is a composition of two involutions (transformations that are their own inverse).



□

5.0.5 Analyticity

To show the Analyticity of $\widetilde{\Phi} := \lim \Phi_\delta$ we need to show that, for $\tau = e^{2\pi i/3}$, $R^\delta = H_1^\delta + H_2^\delta + H_3^\delta$ and $S^\delta = H_1^\delta + \tau H_2^\delta + \tau^2 H_3^\delta$ are holomorphic in the limit. As a result, we will have that $\Phi^\delta = \frac{2}{3}S^\delta + \frac{1}{3}R^\delta$ is also holomorphic.
To do this, we will use Morera's Criterion stating that for any closed contour Γ such that $\oint_\Gamma \widetilde{\Phi} = 0$, then $\widetilde{\Phi}$ is holomorphic. To use this criterion, we will assume that Γ is piecewise smooth and counterclockwise oriented. Now, discretize Γ by a path Γ_δ made of hexagonal edges. Then approximate $\oint_\Gamma \widetilde{\Phi}$ by a sum and we show that the sum goes to 0 as $\delta \to 0$.



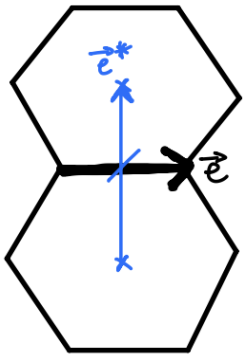
Proof. We consider the following sum for $\overrightarrow{e} = \overrightarrow{xy}$:

$$\sum_{\overrightarrow{e}\in\Gamma_\delta} S^\delta(\overrightarrow{e})\cdot\overrightarrow{e} = \sum_{f\in \mathrm{Int}(\Gamma_\delta)} \sum_{\overrightarrow{e}\in\partial f} S^\delta(\overrightarrow{e})\cdot\overrightarrow{e}. \tag{5.20}$$

In the above we consider ∂f to be counterclockwise, along with $S^\delta(\overrightarrow{e}) = \frac{1}{2}\left(S^\delta(y) - S^\delta(x)\right)$ and $\overrightarrow{e} = y - x$.
For each hexagon f, denoting by $m(\overrightarrow{e}) = \frac{x+y}{2}$ and $c(f)$ the center of f, we can rewrite the previous sum as:

$$\begin{aligned}
\sum_{\overrightarrow{e}\in\partial f} S^\delta(\overrightarrow{e})\cdot\overrightarrow{e} &= \underbrace{-\sum_{\overrightarrow{e}\in\partial f} \partial_{\overrightarrow{e}} S^\delta \cdot m(\overrightarrow{e})}_{\text{Integration by parts formula}} \\
&= -\sum_{\overrightarrow{e}\in\partial f} \partial_{\overrightarrow{e}} S^\delta \cdot (m(\overrightarrow{e}) - c(f)) \\
&= \sum_{\overrightarrow{e}\in\partial f} \partial_{\overrightarrow{e}} S^\delta \cdot \overrightarrow{e}^* \\
&= \sum_{\overrightarrow{e}\in\partial f} \left(\partial^+_{\overrightarrow{e}} S^\delta - \partial^-_{\overrightarrow{e}} S^\delta\right)\cdot\overrightarrow{e}^*,
\end{aligned}$$

where $\overrightarrow{e}^*$ is the dual edge to $\overrightarrow{e}$, rotated by $\frac{\pi}{2}$, viewed as complex numbers.



Now remark that $\partial^-_{\overrightarrow{e}} = \partial^+_{-\overrightarrow{e}}$. Summing over all fs and up to a constant C yields:

$$\sum_{f\in\partial f}\sum_{\overrightarrow{e}\in\partial f} \partial^+_{\overrightarrow{e}} S^\delta\cdot\overrightarrow{e}^* + \sum_{f\in\partial f}\sum_{\overrightarrow{e}\in\partial f} (-\overrightarrow{e}^*)\cdot\partial^+_{\overrightarrow{e}} S^\delta = C\sum_{\overrightarrow{e}\in\mathrm{Int}(\Gamma_\delta)} \left(\partial^+_{\overrightarrow{e}} S^\delta\right)\cdot\overrightarrow{e}^*, \tag{5.21}$$

where the last sum is irrespective of orientation. Furthermore, since we defined S^δ by $S^\delta =$

$H_1^\delta + \tau H_2^\delta + \tau^2 H_3^\delta$ it implies that:

$$\partial^+_{\overrightarrow{e}} S^\delta = \partial^+_{\overrightarrow{e}} H_1^\delta + \tau \partial^+_{\tau^2 \overrightarrow{e}} H_1^\delta + \tau^2 \partial^+_{\tau \overrightarrow{e}} H_1^\delta. \tag{5.22}$$

Hence, applying a change of variable $\overrightarrow{e}^* \to \tau \overrightarrow{e}^*$ and $\overrightarrow{e} \to \tau^2 \overrightarrow{e}^*$ in the second line:

$$\begin{aligned}\sum_{\overrightarrow{e} \in \mathrm{Int}(\Gamma_\delta)} \left(\partial^+_{\overrightarrow{e}} S^\delta\right) \cdot \overrightarrow{e}^* &= \sum_{\overrightarrow{e} \in \mathrm{Int}(\Gamma_\delta)} \overrightarrow{e}^* \cdot \left(\partial^+_{\overrightarrow{e}} H_1^\delta + \tau \partial^+_{\tau^2 \overrightarrow{e}} H_1^\delta + \tau^2 \partial^+_{\tau \overrightarrow{e}} H_1^\delta\right) \\ &= \sum_{\overrightarrow{e} \in \mathrm{Int}(\Gamma_\delta)} \partial^+_{\overrightarrow{e}} H_1^\delta \underbrace{\left(\overrightarrow{e}^* + \tau^2 \overrightarrow{e}^* + \tau \overrightarrow{e}^*\right)}_{=0} = 0.\end{aligned}$$

In this proof we neglected all boundary terms, for which there are $O(1/\delta)$ of them. Since they come with $\overrightarrow{e}^*$ which is $O(\delta)$ and $\partial^+_{\overrightarrow{e}} H_i^\delta$ which is $O(\delta^\alpha)$ by RSW, we can omit them in the calculations. The same proof holds for R^δ.

□

5.1 Holomorphicity & Bijectivity

Now that we have studied precompactness, bounded conditions and analyticity, it remains to look at the possible limits of Φ_{δ_n} as $\delta_n \to 0$ and show that $\widetilde{\Phi} = \Phi$. To prove that $\widetilde{\Phi} = \Phi$ we can simply just show that $\widetilde{\Phi}$ is holomorphic and bijective.

Proposition 5.1.1. *$\widetilde{\Phi}$ is holomorphic and bijective.*

Proof. First note that we already know that the three points a_1, a_2 and a_3 are mapped to 1, $\frac{i\sqrt{3}}{3}$ and $-\frac{i\sqrt{3}}{3}$, and also that Φ is unique.
Now, we need to show that for all $\omega \in \triangleright$ there exists a unique $z \in \Omega$ such that $\Phi(z) = \omega$. To do this, we have to count the number of zeros of $z \mapsto \Phi(z) - \omega$. Recall that to count the number of zeros, we can use the well-know fact from Complex Analysis stating that the number of zeros of a holomorphic function on Ω with $\partial\Omega = \gamma$ is:

$$\frac{1}{2\pi i} \oint_\gamma \frac{f'}{f} dz. \tag{5.23}$$

Hence, consider that the boundary $\partial\Omega$ is oriented counterclockwise. We use the above fact and a change of variable to find:

$$\frac{1}{2\pi i} \oint_{\partial\Omega} \frac{\widetilde{\Phi}'(z)}{\widetilde{\Phi}(z) - \omega} dz = \frac{1}{2\pi i} \int_{\partial\triangleright} \frac{d\xi}{\xi - \omega} = \begin{cases} 1 & \text{if } \omega \in \triangleright \\ 0 & \text{if } \omega \in \mathbb{C} \backslash \triangleright \end{cases}. \tag{5.24}$$

□

Chapter 6

Ising Model

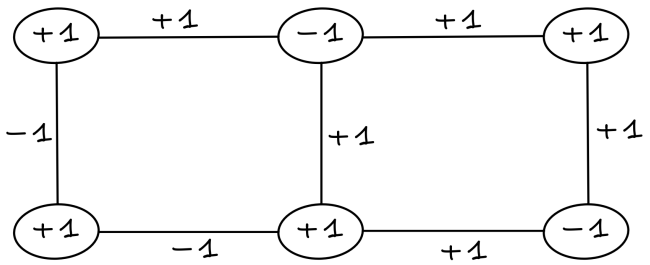
The Ising Model is a random assignment of spins $\{\sigma_i\}_{i\in V}$ to the vertices of a graph $G=(V,E)$, where $\sigma_i \in \{\pm 1\}$. We have that the probability of spin configuration is:

$$\mathbb{P}_\beta(\sigma) \propto \exp\left(\beta \sum_{(i,j)\in E} \sigma_i\sigma_j\right). \tag{6.1}$$

We also usually write,

$$\mathbb{P}(\sigma) = \frac{1}{Z} e^{-\beta H(\sigma)}, \tag{6.2}$$

where $H(\sigma) = -\sum_{(i,j)\in E}\sigma_i\sigma_j$ and $\beta = \frac{1}{T} > 0$ a parameter. Note that we also consider the so-called *partition function* $Z = \sum_{\sigma\in\{\pm1\}^V} e^{-\beta H(\sigma)}$. For example in the following, $H(\sigma) = 3$:



We can now have several questions about this model: What happens on a large graph? Is there a global alignment phenomenon? If $\Omega \subseteq \mathbb{R}^d$ and $\Omega_\delta = \Omega \cap \delta\mathbb{Z}^d$ is a discretization, what do we see as $\delta \to 0$? Note that this may sometimes depend on β.

6.1 Simulation

We first present the Metropolis Algorithm, which allows us to simulate the Ising Model.

Proposition 6.1.1 (Metropolis Algorithm)**.** *We consider the following algorithm:*
(1) Start from an arbitrary configuration σ.
(2) Pick a random vertex $x \in V$ and compute the energy H before flipping the value of σ_x and after.
(3) If $H_{after} < H_{before}$, keep the new configuration otherwise keep it with probability

$$\frac{\mathbb{P}_\beta(\sigma_{after})}{\mathbb{P}_\beta(\sigma_{before})} = e^{-\beta\left(H(\sigma_{after}) - H(\sigma_{before})\right)}. \tag{6.3}$$

(4) Go back to (2).
(5) Repeat for many steps.

Note that $(H(\sigma_{\text{after}}) - H(\sigma_{\text{before}}))$ is an easy quantity to compute. Now why does this work? We are performing a random walk on the space of configurations (two configurations are neighbours if they differ by one spin). So we are considering a Markov Chain on $\{\pm 1\}^V$ with matrix,

$$M_{\sigma,\sigma'} = \frac{1}{|V|}\min\left\{1, e^{-\beta(H(\sigma')-H(\sigma))}\right\}, \qquad \text{if } \sigma \sim \sigma'. \tag{6.4}$$

The measure we are sampling from after n steps if we started from μ is μM^n, where μ is a line vector. If as $n \to \infty$ there is only one μ_∞ such that $\mu_\infty M = \mu_\infty$ and $\mu_{\text{Ising}} M = \mu_{\text{Ising}}$ then $\mu_\infty = \mu_{\text{Ising}}$.

Remark 6.1.2. *We can make the following remarks:*
(1) $\mu_\infty M = \mu_\infty$ if and only if $M^T \mu_\infty^T = \mu_\infty^T$.
(2) M^T has non-negative entries and $(M^T)^{|V|}$ has strictly positive entries.
(3) Perron-Frobenius on $(M^T)^{|V|}$ has a unique leading positive eigenvector of eigenvalues ≤ 1 (since we have Markov chains).
(4) M^T also (if it had more than 1, then $(M^T)^{|V|}$ would also).
(5) $\mu_{Ising,\beta} M =_{(*)} \mu_{Ising,\beta}$ because of detailed balance equation, i.e. for all σ, σ',

$$\mu_{Ising,\beta}(\sigma) M_{\sigma,\sigma'} =_{(**)} \mu_{Ising,\beta}(\sigma') M_{\sigma',\sigma}. \tag{6.5}$$

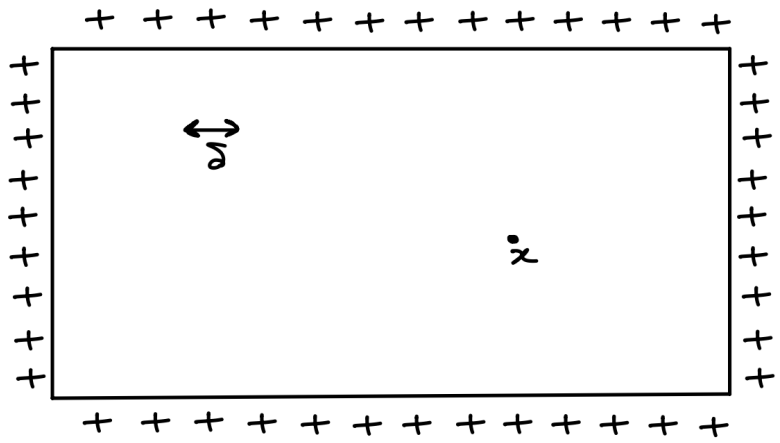
$(**)$ *implies* $(*)$ *as a result of summing over σ and getting $(\mu M)_{\sigma'} = \mu_{\sigma'}$.*

The Ising Model is a natural measure if we want:
- Nearest neighbour interactions (conditionally on neighbours, spin at a vertex is independent of the other spins).
- The interaction strength is the same for every edge.
- Spins take values in $\{\pm 1\}$ and $+1$ is symmetric with respect to -1.

The entropy of a discrete probability measure is $\mathcal{H}(\mathbb{P}) = -\sum_\omega \log(\mathbb{P}(\omega))\mathbb{P}(\omega)$. If we have a spin configuration and we know its energy $H(\sigma) = -\sum_{(i,j)\in E} \sigma_i \sigma_j$, the maximum entropy distributes among the σs of a given H, and it should be uniform. Now, if we only know the value of $\mathbb{E}(H) = \sum_\sigma H(\sigma)\mathbb{P}(\sigma)$, the maximal entropy distribution with this constraint will be $\mathbb{P}(\sigma) \propto e^{-\beta H}$ for some $\beta > 0$ (using Lagrange multipliers).

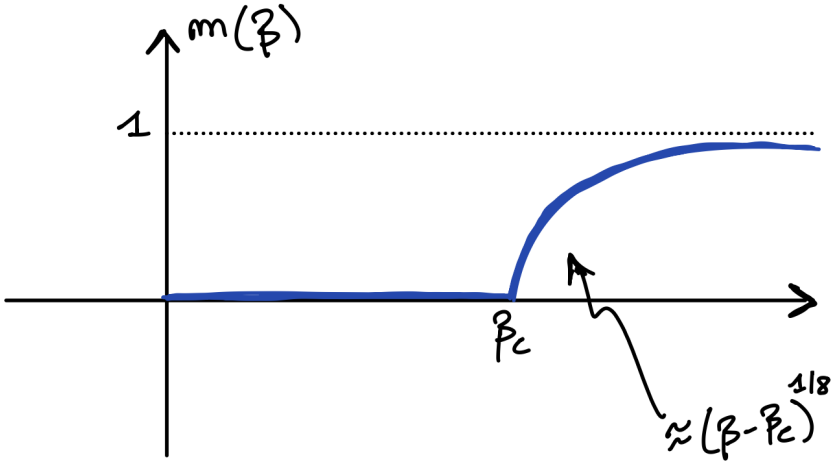
Metropolis dynamics are not the only updates that satisfies detailed balance with respect to $\mathbb{P}_{\text{Ising}}$. Another useful one is the so-called *Heat Bath* or *Glauber Dynamics*. Instead of proposing a flip, pick a vertex x and resample σ_x according to $\mathbb{P}_{\text{Ising}}$ knowing the σ_y for $y \sim x$.

Let's now look at the large-scale Ising Model. Consider, for example, a domain $\Omega \subset \mathbb{R}^2$ and $\Omega_\delta = \Omega \cap \delta\mathbb{Z}^2$ a discretization. Consider also that we take the $+$ boundary conditions (take spins on $\partial\Omega_\delta$ to be $+1$).



Given $(\sigma_x)_{x\in\Omega_\delta}$ and $H(\sigma) = -\sum_{(x,y)\in E\cup\partial E}\sigma_x\sigma_y$ (∂E represents the edges between Ω_δ and $\partial\Omega_\delta$), what can we say about the limit of $\mathbb{E}(\sigma_x)$ as $\delta \to 0$? (We omit the proof of this theorem)

Theorem 6.1.3. *For $\xi(\beta) > 0$, $\beta_c = \frac{1}{2}\ln(\sqrt{2}+1)$, the limits l are given by:*
→ *If $\beta < \beta_c$, the limit is $l = \mathcal{O}\left(e^{-\xi(\beta)\delta^{-1}}\right)$.*
→ *If $\beta > \beta_c$ and $m(\beta) > 0$, the limit is $l = m(\beta) + \mathcal{O}\left(e^{-\xi(\beta)\delta^{-1}}\right)$.*
→ *If $\beta = \beta_c$ and for some constant C_Ω, the limit is $l = C_\Omega\delta^{\frac{1}{8}}$.*

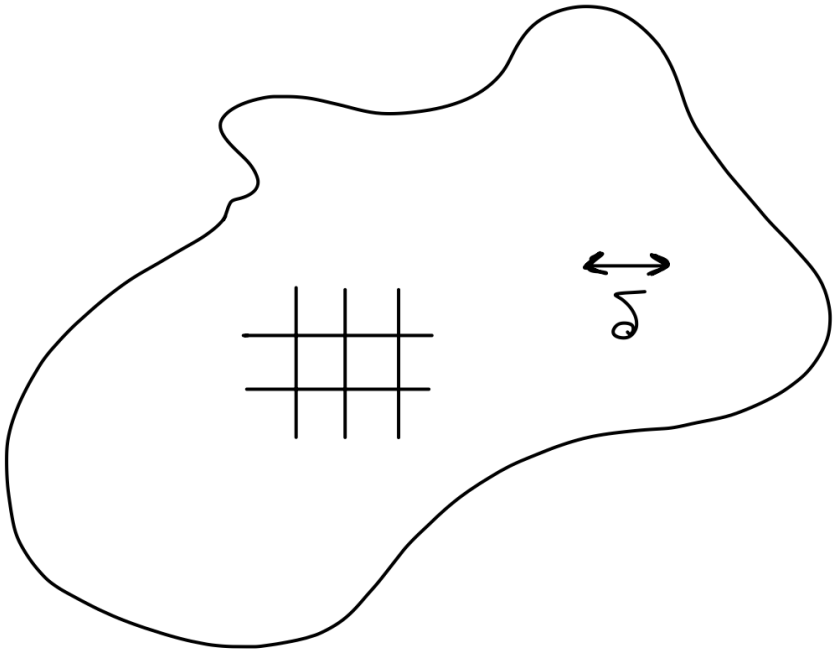


Remark 6.1.4. *In dimension $d = 1$, $\beta_c = \infty$, whereas in dimension $d \geq 2$, $\beta_c < \infty$.*

For now, can we prove that $\mathbb{E}_{\Omega_\delta}(\sigma_x)$ is monotonous with respect to boundary conditions, i.e. if we fix $\sigma|_{\partial\Omega} = f$, $\sigma|_{\partial\Omega} = g$ and $f \leq g$ then $\mathbb{E}_f(\sigma_x) \leq \mathbb{E}_g(\sigma_y)$?
The above theorem implies that with $+$ boundary conditions, because $\mathbb{E}(\sigma_x)$ decreases as δ gets smaller, an Ising Model in a larger box can always be viewed as an Ising Model in a smaller box with the appropriate (random) boundary condition (inside and outside are conditionally independent given the boundary). Hence the answer is no, boundary conditions gives a higher expectation than $+1$ on the smaller box.

6.2 High & Low Temperature Representations

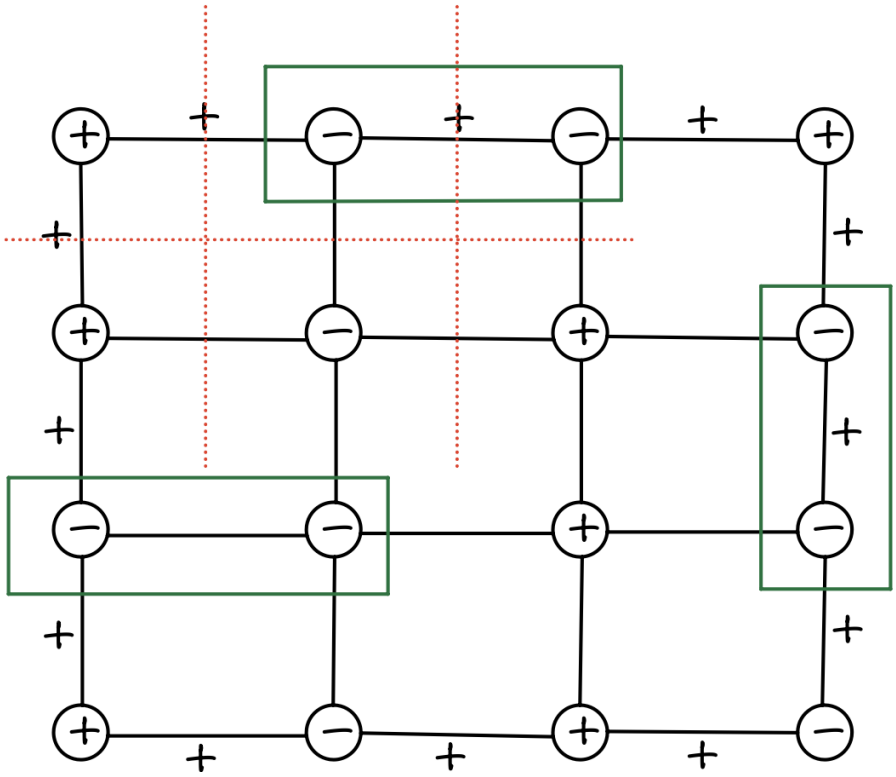
In this section, we fix β and look at the behaviour of the system when $\delta \to 0$.



We will show that, in 2 dimensions, for β small we have no long-range order while for β large we do have a long-range order.

Theorem 6.2.1 (Peierls' Theorem)**.** *In 2 dimensions, there is a long-range order for β large enough.*

Consider the dual graph Ω_δ^* to Ω_δ. If we look at the Ising Model with + boundary conditions on Ω_δ (spins on $\partial\Omega_\delta$ are equal to $+1$), the σ configurations can be described by loop configurations $\omega < \mathcal{E}(\Omega_\delta^*)$, with an edge $e^* \in \omega$ if and only if $\sigma_x \neq \sigma_y$ for $e = (x, y)$.



Looking at $\mathbb{P}(\omega)$ we get:

$$\mathbb{P}(\omega) \propto e^{-\beta H(\sigma)} \propto e^{-\beta|\omega|+(|\mathcal{E}(\Omega_\delta)|-\beta|\omega|)} \propto e^{-2\beta|\omega|}, \tag{6.6}$$

where $|\omega|$ is the number of edges in ω, $(-\beta|\omega|)$ represents the pairs of spins that are different and $(|\mathcal{E}(\Omega_\delta)|-\beta|\omega|)$ represents the pairs of spins that are the same. This is the so-called Low Temperature Representation, and it is particularly useful at large β.

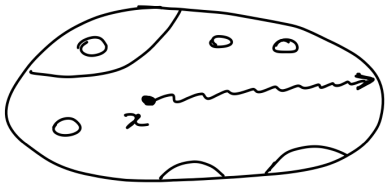
We now shift our interest to the High Temperature Representation, where we start with the Ising partition function:

$$\begin{aligned}
Z_{\Omega_\delta} = \sum_{\sigma\in\{\pm1\}^V} e^{\beta\sum_{i\sim j}\sigma_i\sigma_j} &= \sum_{\sigma\in\{\pm1\}^V}\prod_{i\sim j} e^{\beta\sigma_i\sigma_j} \\
&= \sum_{\sigma\in\{\pm1\}^V}\prod_{i\sim j}(\cosh(\beta\sigma_i\sigma_j)+\sinh(\beta\sigma_i\sigma_j)) \\
&= \sum_{\sigma\in\{\pm1\}^V}\prod_{i\sim j}(\cosh(\beta)+\sigma_i\sigma_j\sinh(\beta)) \\
&= \sum_{\sigma\in\{\pm1\}^V}(\cosh(\beta))^{|\mathcal{E}|}\prod_{i\sim j}(1+\sigma_i\sigma_j\tanh(\beta)) \\
&= (\cosh(\beta))^{|\mathcal{E}|}\sum_{\sigma\in\{\pm1\}^V}\sum_{\gamma\subset\mathcal{E}}\prod_{(i,j)\in\gamma}\sigma_i\sigma_j\tanh(\beta) \\
&= (\cosh(\beta))^{|\mathcal{E}|}\sum_{\gamma\subset\mathcal{E}}(\tanh(\beta))^{|\gamma|}\sum_{\sigma\in\{\pm1\}^V}\prod_{(i,j)\in\gamma}\sigma_i\sigma_j \\
&= (\cosh(\beta))^{|\mathcal{E}|}\sum_{\gamma\subset\mathcal{E}}(\tanh(\beta))^{|\gamma|}\,2^{|V|}\mathbb{1}_{\{\text{each } x \text{ appears in an even number of edges of } \gamma\}}.
\end{aligned}$$

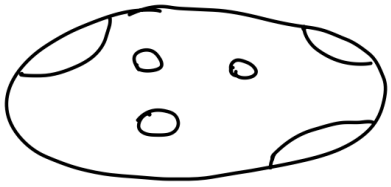
At this point it looks like there is a duality between Ising on Ω_δ^* with $+$ boundary conditions and β, and Ising on Ω_δ with free boundary conditions and β^* (with $H(\beta^*)\propto e^{-2\beta}$).
Let us now look at $\mathbb{E}^+_{\Omega_\delta}(\sigma_x)$ (i.e. with $+$ boundary conditions) via high temperature representation techniques. Including boundary edges in $\sum_{i\sim j}\sigma_i\sigma_j$ we have:

$$\mathbb{E}^+_{\Omega_\delta}(\sigma_x) = \frac{\sum_{\sigma\in\{\pm1\}^{V(\Omega_\delta)}}\sigma_x e^{\beta\sum_{i\sim j}\sigma_i\sigma_j}}{\sum_{\sigma\in\{\pm1\}^{V(\Omega_\delta)}}e^{\beta\sum_{i\sim j}\sigma_i\sigma_j}} = \ldots\ldots = \frac{\sum_{\widetilde{\gamma}\subset\mathcal{E}}(\tanh(\beta))^{|\widetilde{\gamma}|}\sum_{\sigma\in\{\pm1\}^V}\sigma_x\prod_{i\sim j\in\widetilde{\gamma}}(\sigma_i\sigma_j)}{\sum_{\gamma\subset\mathcal{E}}(\tanh(\beta))^{|\gamma|}\sum_{\sigma\in\{\pm1\}^V}\prod_{i\sim j\in\gamma}(\sigma_i\sigma_j)}.$$

Above, we considered $\widetilde{\gamma}$ with loops + arcs + path x:



and γ with loops + arcs:



The term highlighted in green is given by:

$$\sum_{\sigma\in\{\pm1\}^V}\sigma_x\prod_{i\sim j\in\widetilde{\gamma}}(\sigma_i\sigma_j) = \begin{cases} 2^{|V|}, & \text{if } x \text{ belongs to an odd number of edges, all other vertices} \\ & \text{belong to an even number of edges except boundary ones that are free.} \\ 0, & \text{otherwise.} \end{cases}$$

Hence we obtain the following simplification:

$$\mathbb{E}^+_{\Omega_\delta}(\sigma_x) = \frac{\sum_{\widetilde{\gamma}} (\tanh(\beta))^{|\widetilde{\gamma}|}}{\sum_{\gamma} (\tanh(\beta))^{|\gamma|}}. \tag{6.7}$$

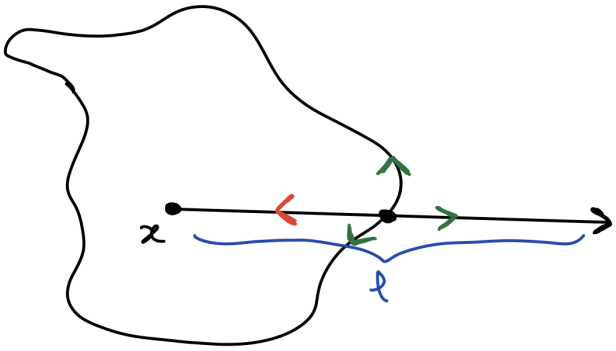
Proof. (Peierls' Theorem)
We need to show that, if β is large enough,

$$\limsup_{\delta \to 0} \mathbb{E}^+_{\Omega_\delta}(\sigma_x) > 0. \tag{6.8}$$

Let us prove that in the low temperature representation the probability that no loop surrounding x is strictly greater than $\frac{1}{2}$ as $\delta \to 0$, i.e. that the probability that there exists > 1 loop is strictly less than $\frac{1}{2}$.

$$(*) := \frac{\sum_{\substack{\widetilde{\omega} \text{ loops, } \widetilde{\omega} \in \epsilon(\Omega^*_\delta) \\ \omega \text{ has a loop } L \text{ surrounding } x}} e^{-2\beta|\widetilde{\omega}|}}{\sum_{\omega \text{ loops, } \omega \in \epsilon(\Omega_\delta)} e^{-2\beta|\omega|}}.$$

Now, with the picture below, we see that the number of loops of length l in $\mathcal{E}(\Omega_\delta)$ that surround x is less than or equal to $l3^l$. This is due to the fact that we either cannot go back or move in 3 other directions.



Finally:

$$\begin{aligned}
(*) &= \frac{\sum_{l=4}^{\infty} \sum_{L \text{ a loop around } x \text{ of length } l} \left(e^{-2\beta l}\right) \sum_{\omega' \text{ loops in } \mathcal{E}(\Omega_\delta)\setminus L} \left(e^{-2\beta|\omega'|}\right)}{\sum_{\omega} \left(e^{-2\beta|\omega|}\right)} \\
&= \sum_{l=4}^{\infty} \sum_{L \text{ a loop around } x \text{ of length } l} \left(e^{-2\beta l}\right) \underbrace{\left(\frac{\sum_{\omega' \text{ loops in } \mathcal{E}(\Omega_\delta)\setminus L} \left(e^{-2\beta|\omega'|}\right)}{\sum_{\omega} \left(e^{-2\beta|\omega|}\right)} \right)}_{\leq 1} \\
&\leq \sum_{l=4}^{\infty} l3^l e^{-2\beta l} \xrightarrow{\beta \to \infty} 0.
\end{aligned}$$

Therefore, there exists a $\beta > 0$ such that the sum is $< \frac{1}{2}$.

□

Let's now summarize what we have seen so far. In 2 dimensions:
(1) At low temperature (high β) expansion, we have spin configurations with $\mathbb{P}(\omega) \propto e^{-2\beta|\omega|}$ if and only if ($\Longleftrightarrow$) we have a collection of loops ω on Ω^*_δ (say with respect to + boundary conditions).

(2) At high temperature (low β) expansion,

$$Z = 2^{|V|}(\cosh(\beta))^{|E|} \sum_{\gamma:\text{ loops on } \Omega_\delta} (\tanh(\beta))^{|\gamma|}. \tag{6.9}$$

We have a duality between {Ising on Ω_δ^* with respect to + boundary conditions at inverse temperature β} and {Ising on Ω_δ with respect to free boundary conditions at inverse temperature β^* such that $\tanh(\beta^*) = e^{-2\beta}$}.

Theorem 6.2.2 (Kramers-Wannier Duality)**.** *Considering the partition function of the Ising Model with free boundary conditions at arbitrary inverse temperature β on the graph $G = (V, E)$, $Z^{\varnothing}_{\Omega,\beta}$, and the partition function of the Ising Model with + boundary conditions at arbitrary inverse temperature β^* on the dual graph $G^* = (V^*, E^*)$, $Z^{+}_{\Omega^*,\beta^*}$, we have the following duality:*

$$e^{-\beta^*|E^*|} Z^{+}_{\Omega^*,\beta^*} = \cosh(\beta)^{-|E|} 2^{-|V|} Z^{\varnothing}_{\Omega,\beta}. \tag{6.10}$$

Proof. First recall that with the given setup, the parameter β^* satisfies the relation $e^{-2\beta^*} = \tanh(\beta)$ if and only if $\sinh(\beta^*)\sinh(\beta) = 1$, which in particular implies that $\beta^*(\beta)$ is decreasing in β and $\beta^*(\beta) = \beta \iff \beta = \beta_c := \frac{1}{2}\log(\sqrt{2}+1)$. Note also that V^* is the set of faces of Ω and E^* the set of dual edges (pairs of faces sharing an edge in E).
Now using the high temperature expansion of the partition function $Z^{\varnothing}_{\Omega,\beta}$ and the low temperature expansion of $Z^{+}_{\Omega^*,\beta^*}$ we get:

$$Z^{+}_{\Omega^*,\beta^*} = e^{\beta^*|E^*|} \sum_{\sigma\in\{\pm1\}^{\Omega^*}} \prod_{\gamma\in C(\sigma)} e^{-2\beta^*|\gamma|}. \tag{6.11}$$

$$Z^{\varnothing}_{\Omega,\beta} = \cosh(\beta)^{|E|} 2^{|V|} \sum_{\mathcal{E}\in C} \tanh(\beta)^{|\mathcal{E}|}. \tag{6.12}$$

Here, $C(\sigma)$ is the set of loops in Ω defined by the edges connecting spins of σ with different values and where C is the collection of sets of loops in Ω. Thus $\{C(\sigma)|\sigma \in \{\pm1\}^{\Omega^*}\} = \{\mathcal{E}|\mathcal{E} \in C\}$ since both sides describe the collection of sets of loops on Ω. We can the rewrite the high temperature expansion as $Z^{\varnothing}_{\Omega,\beta} = \cosh(\beta)^{|E|}2^{|V|}\sum_{\sigma\in\{\pm1\}^{\Omega^*}}\prod_{\gamma\in C(\sigma)} e^{-2\beta^*|\gamma|}$. Hence we obtain the required duality:

$$e^{-\beta^*|E^*|} Z^{+}_{\Omega^*,\beta^*} = \cosh(\beta)^{-|E|} 2^{-|V|} Z^{\varnothing}_{\Omega,\beta}. \tag{6.13}$$

□

Peierls' argument for β large enough gave us,

$$\liminf_{\delta\to 0} \mathbb{E}^{+}_{\Omega_\delta}(\sigma_x) > 0, \tag{6.14}$$

where the idea was to show that the probability that there is a number > 0 of loops around x is $\leq Cst < \frac{1}{2}$ uniformly with respect to δ.
We now state and prove the dual Peierls' argument (using high temperature expansion).

Theorem 6.2.3 (Dual Peierls' Theorem)**.** *In 2 dimensions, when β is small enough:*

$$\mathbb{E}^{+}_{\Omega_\delta}(\sigma_x) \xrightarrow{\delta\to 0} 0. \tag{6.15}$$

Proof. As before, we consider γ to be loops + arcs + path $x \to \partial\Omega_\delta$ and $\widetilde{\gamma}$ to be loops + arcs. Now:

$$\begin{aligned}
\mathbb{E}^{+}_{\Omega_\delta}(\sigma_x) &= \frac{\sum_{\gamma}(\tanh(\beta))^{|\gamma|}}{\sum_{\widetilde{\gamma}}(\tanh(\beta))^{|\widetilde{\gamma}|}} \\
&= \frac{\sum_{\pi\text{: path from } x\to\partial\Omega_\delta}(\tanh(\beta))^{|\pi|}\sum_{\gamma'\text{: loops + arcs staying in }\Omega_\delta\setminus\pi}(\tanh(\beta))^{|\gamma'|}}{\sum_{\widetilde{\gamma}}(\tanh(\beta))^{|\widetilde{\gamma}|}} \\
&= \sum_{\pi\text{: path from } x\to\partial\Omega_\delta}(\tanh(\beta))^{|\pi|}\underbrace{\left(\frac{\sum_{\gamma'\text{: loops + arcs staying in }\Omega_\delta\setminus\pi}(\tanh(\beta))^{|\gamma'|}}{\sum_{\widetilde{\gamma}}(\tanh(\beta))^{|\widetilde{\gamma}|}}\right)}_{\leq 1} \\
&\leq \sum_{\pi\text{: path from } x\to\partial\Omega_\delta}(\tanh(\beta))^{|\pi|} \\
&\leq \sum_{l=\lfloor\frac{c}{\delta}\rfloor}^{\infty}(\tanh(\beta))^{l}\cdot 4^{l}
\end{aligned}$$

where we have used the facts that π needs at least $\lfloor\frac{c}{\delta}\rfloor$ edges to go to $\partial\Omega_\delta$ (for some constant c), and that 4^l is a bound on the number of paths of length l. Therefore:

$$\mathbb{E}^{+}_{\Omega_\delta}(\sigma_x) \leq \sum_{l=\lfloor\frac{c}{\delta}\rfloor}^{\infty}(\tanh(\beta))^{l}\cdot 4^{l} = \frac{1}{1-4\tanh(\beta)}|4\tanh(\beta)|^{\lfloor\frac{c}{\delta}\rfloor} \xrightarrow{\delta\to 0} 0, \quad \text{if } 4\tanh(\beta)<1.$$

□

Proposition 6.2.4. *For every inverse temperature $\beta \in (0,\infty)$ we have:*

$$\mathbb{E}^{\varnothing}_{\Omega,\beta}(\sigma_x\sigma_y) > 0. \tag{6.16}$$

Proof. In the high temperature expansion for the partition function and the intersection of two spins is:

$$Z^{\varnothing}_{\Omega,\beta} = \sum_{\sigma} e^{-\beta H(\sigma)} = \cosh(\beta)^{|E|}2^{|V|}\sum_{\mathcal{E}\in C}\tanh(\beta)^{|\mathcal{E}|}. \tag{6.17}$$

$$\mathbb{E}^{\varnothing}_{\Omega,\beta}(\sigma_x\sigma_y) = \frac{1}{Z^{\varnothing}_{\Omega,\beta}}\sum_{\sigma}\sigma_x\sigma_y e^{-\beta H(\sigma)} = \frac{1}{Z^{\varnothing}_{\Omega,\beta}}\cosh(\beta)^{|E|}2^{|V|}\sum_{\mathcal{E}\in C_{x,y}}\tanh(\beta)^{|\mathcal{E}|}. \tag{6.18}$$

We have that C is the collection of edge sets such that every vertex is incident to an even number of edges and $C_{x,y}$ is the collection of edge sets where every vertex is incident to an even number of edges except for x and y which are incident to an odd number of edges (i.e. C is the set of set

of loops and $C_{x,y}$ is the set of loops with a path $x \longleftrightarrow y$). Therefore:

$$\mathbb{E}^{\varnothing}_{\Omega,\beta}(\sigma_x\sigma_y) = \frac{\sum_{\mathcal{E}\in C_{x,y}} \tanh(\beta)^{|\mathcal{E}|}}{\sum_{\mathcal{E}\in C} \tanh(\beta)^{|\mathcal{E}|}} > 0, \tag{6.19}$$

since every term in $\sum_{\mathcal{E}\in C_{x,y}} \tanh(\beta)^{|\mathcal{E}|}$ and $\sum_{\mathcal{E}\in C} \tanh(\beta)^{|\mathcal{E}|}$ is positive and the sum in the numerator is non-empty.

□

Proposition 6.2.5. *We consider the Ising model with + boundary conditions on the square lattice inside the open unit disc $D \subset \mathbb{R}^2$. We denote by D_δ the discretization $D \cap \delta\mathbb{Z}^2$. There exists an inverse temperature $\beta \in (0,\infty)$ large enough such that:*

$$\liminf_{\delta\to 0} \mathbb{E}^{\beta}_{D_\delta,+}(\sigma_{(0,0)}) \geq 0.99. \tag{6.20}$$

Proof. We have that:

$$\mathbb{E}^{\beta}_{D_\delta,+}(\sigma_{(0,0)}) = \mathbb{P}^{\beta}_{D_\delta,+}(\sigma_{(0,0)} = 1) - \mathbb{P}^{\beta}_{D_\delta,+}(\sigma_{(0,0)} = -1) = 1 - 2\mathbb{P}^{\beta}_{D_\delta,+}(\sigma_{(0,0)} = -1). \tag{6.21}$$

Let us then fix $\delta > 0$ such that $\mathbb{P}^{\beta}_{D_\delta,+}(\sigma_{(0,0)} = -1) \leq \epsilon(\beta)$ where $\epsilon(\beta) \to 0$ as $\beta \to \infty$ is a function independent of δ. Hence we will have that $\mathbb{E}^{\beta}_{D_\delta,+}(\sigma_{(0,0)}) \geq 1 - 2\epsilon(\beta)$, and to get the desired estimate we will require β large enough so that $2\epsilon(\beta) < 0.01$.
Recall the partition function $Z_{D_\delta,+} = \sum_{\sigma\in\{\pm1\}^{D_\delta,+}} e^{\beta\sum_{\{x,y\}\in\mathcal{E}}\sigma_x\sigma_y}$. We can express the Hamiltonian as:

$$H_{D_\delta,+}(\sigma) = -\sum_{\{x,y\}\in\mathcal{E}} \sigma_x\sigma_y = -|\mathcal{E}| + \sum_{\{x,y\}\in\mathcal{E}} (1 - \sigma_x\sigma_y) = -|\mathcal{E}| + 2|\{\{x,y\}\in\mathcal{E} | \sigma_x \neq \sigma_y\}|. \tag{6.22}$$

We have seen that each configuration σ defines a set of loops in the dual graph of D_δ, each loop edge corresponds to an original edge between vertices whose spins differ; we denote this set of loops as $C(\sigma)$. We can thus rewrite the last sum as $H_{D_\delta,+}(\sigma) = -|\mathcal{E}| + 2\sum_{\gamma\in C(\sigma)}|\gamma|$. Thus the partition function is:

$$Z_{D_\delta,+} = \sum_{\sigma\in\{\pm1\}^{D_\delta,+}} e^{\beta|\mathcal{E}| - 2\beta\sum_{\gamma\in C(\sigma)}|\gamma|} = e^{\beta|\mathcal{E}|} \sum_{\sigma\in\{\pm1\}^{D_\delta,+}} \prod_{\gamma\in C(\sigma)} e^{-2\beta|\gamma|}. \tag{6.23}$$

Notice that when we consider the probability of a particular configuration σ using this expression, the term $e^{\beta|\mathcal{E}|}$ cancels out in the numerator and denominator, so we can disregard it.
Now note that in order for $\sigma_{(0,0)} = -1$, the vertex $(0,0)$ has to be surrounded by an odd number of loops. This in particular implies, that there needs to exist at least one loop surrounding $(0,0)$. Hence we have that:

$$\begin{aligned}\mathbb{P}^{\beta}_{D_\delta,+}(\sigma_{(0,0)} = -1) \leq \mathbb{P}^{\beta}_{D_\delta,+}(\exists\gamma^* \in C(\sigma) \text{ a loop surrounding } (0,0)) &\leq \sum_{\gamma^*} \mathbb{P}^{\beta}_{D_\delta,+}(\gamma^*) \\ &= \sum_{\gamma^*} \sum_{\sigma:\gamma^*\in C(\sigma)} \mathbb{P}^{\beta}_{D_\delta,+}(\sigma).\end{aligned}$$

Using (6.23) we can rewrite the above expression using contours:

$$\sum_{\gamma^*}\sum_{\sigma:\gamma^*\in C(\sigma)}\mathbb{P}^{\beta}_{D_\delta,+}(\sigma)=\sum_{\gamma^*}\frac{\sum_{\sigma:\gamma^*\in C(\sigma)}\prod_{\gamma\in C(\sigma)}e^{-2\beta|\gamma|}}{\sum_{\sigma'}\prod_{\gamma'\in C(\sigma')}e^{-2\beta|\gamma'|}}$$
$$=\sum_{\gamma^*}e^{-2\beta|\gamma^*|}\underbrace{\frac{\sum_{\sigma:\gamma^*\in C(\sigma)}\prod_{\gamma\in C(\sigma)\setminus\{\gamma^*\}}e^{-2\beta|\gamma|}}{\sum_{\sigma'}\prod_{\gamma'\in C(\sigma')}e^{-2\beta|\gamma'|}}}_{\leq 1\ (*)}\leq\sum_{\gamma^*}e^{-2\beta|\gamma^*|}.$$

For $(*)$, we will show that the numerator is less than or equal to the denominator as follows: for every configuration σ that contains the loop γ^* we construct a new configuration σ' such that $\prod_{\gamma\in C(\sigma)\setminus\{\gamma^*\}}e^{-2\beta|\gamma|}=\prod_{\gamma'\in C(\sigma')}e^{-2\beta|\gamma'|}$. In other words, we want σ' to have the exact same loops as σ, just without γ^*. To construct σ', simply take σ and flip all the spins in the interior of the γ^* loop; it is not hard to see that the resulting configuration σ' will have all the loops as σ, except for γ^*. Moreover, from the construction, it is apparent that the mapping $\sigma\mapsto\sigma'$ is injective. Thus, each product $\prod_{\gamma\in C(\sigma)\setminus\{\gamma^*\}}e^{-2\beta|\gamma|}$ in the numerator is also present in the denominator as $\prod_{\gamma'\in C(\sigma')}e^{-2\beta|\gamma'|}$ which concludes the proof of $(*)$.

Now, it remains to estimate the number of loops around $(0,0)$ with a fixed length l. We just have to find a bound on the number of loops of length l which surround $(0,0)$. Yet since such a loop surrounds $(0,0)$ it must cross the x-axis at a point which is at a maximal distance from 0 of order l (i.e. up to a multiplicative constant). Thus we can consider that the loop starts at a point which is at a maximal distance from 0 of order l (there are l such points) and then it is of length l: there is still a maximum of 4^l choices of loops once we have chosen its starting points. Thus there exists a constant C >0 such that $|\{\text{loops of length } l \text{ surrounding } (0,0) \text{ in the dual graph}\}|\leq Cl4^l$. Now,

$$\mathbb{P}^{\beta}_{D_\delta,+}(\sigma_{(0,0)}=-1)\leq\sum_{\gamma^*}e^{-2\beta|\gamma^*|}\leq C\sum_{l\geq 1}l4^le^{-2\beta l}\xrightarrow{\beta\to\infty}0. \tag{6.24}$$

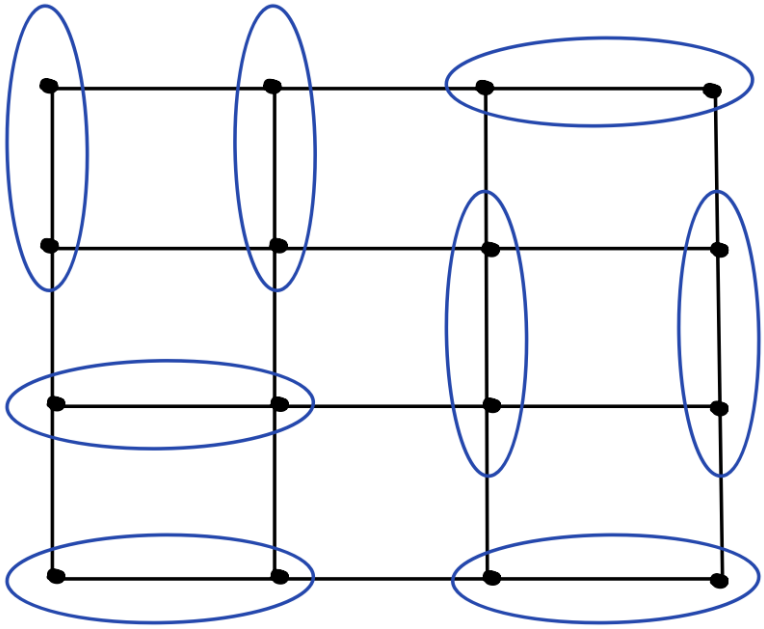
We have shown that $\mathbb{P}^{\beta}_{D_\delta,+}(\sigma_{(0,0)}=-1)\leq\epsilon(\beta)$ where $\epsilon(\beta)\to 0$ as $\beta\to\infty$ is a function independent of δ, hence concluding the proof.

□

Chapter 7

Dimer Model

In this section we present the Dimer Model/Perfect Matching Model/Domino Tiling Model. For a graph $G = (V, E)$, a perfect matching is a sub-collection $\omega \subset E$ such that every vertex belongs to an edge.



One of the classic questions in this domain is: can you cover a chessboard of 62 cells with dominos? Answer: No! If we color the squares in a chessboard fashion, 32 black squares and 30 white squares will cover it. Since each domino would cover 1 white square and 1 black square then the covering is impossible.

Our objective in this chapter is to obtain a nice formula for the number of domino tilings of an $m \times n$ chessboard. Note that if both m and n are odd then this number is (obviously) zero.

Theorem 7.0.1 (Kasteleyn - 1965)**.** *Setting $N = \frac{mn}{2}$, the number of domino tilings of a $m \times n$ chessboard is given by:*

$$\sqrt{\prod_{k=1}^{m}\prod_{l=1}^{n}\left(2\cos\left(\frac{\pi k}{m+1}\right)+2i\cos\left(\frac{\pi l}{n+1}\right)\right)}. \tag{7.1}$$

Definition 7.0.2. *The **Adjacency matrix** $\widetilde{A}$ of a graph $G=(V,E)$ is the $|V|\times|V|$ matrix indexed by vertices such that $\widetilde{A}_{ij}=\delta_{i\sim j}$.*

Definition 7.0.3. *A graph is said to be **bipartite** if its vertex set can be written as the disjoint union $V=B\sqcup W$ where $v\sim w$ in G implies $v\in B$ and $w\in W$ or $w\in B$ and $v\in W$.*

Of course, for a bipartite G, for it to be dimer-tilable we require $|B|=|W|=N$.

Definition 7.0.4. *For a bipartite graph, the **Reduced Adjacency matrix** is the $|B|\times|W|$ matrix indexed by $A_{bw}=\delta_{b\sim w}$.*

Remark 7.0.5. *One can label vertices such that $\widetilde{A}=\begin{pmatrix}0 & A\\ A^T & 0\end{pmatrix}$.*

Lemma 7.0.6. *The number of domino tilings of G is equal to the permanent $\mathrm{Per}(A)$ of A, where $\mathrm{Per}(A):=\sum_{\sigma\in S_N}a_{1\sigma(1)}...a_{N\sigma(N)}$.*

Proof. We have:

$$\mathrm{Per}(A)=\sum_{\sigma\in\mathrm{Per}(B)}\underbrace{\prod_{b\in B}a_{b\sigma(b)}}_{=1\Longleftrightarrow b\sim\sigma(b),\forall b\in B}=\sum_{T:\text{ dimer tiling}}1=\text{Number of domino tilings of }G.$$

□

The problem is that the permanent is very difficult to compute, so we would like a determinant instead.

Definition 7.0.7. *We define the **Reduced Kasteleyn matrix** K as:*

$$k_{bw}:=\begin{cases}1, & \text{if } (b,w) \text{ is a horizontal edge.}\\ i, & \text{if } (b,w) \text{ is a vertical edge.}\\ 0, & \text{otherwise.}\end{cases}\tag{7.2}$$

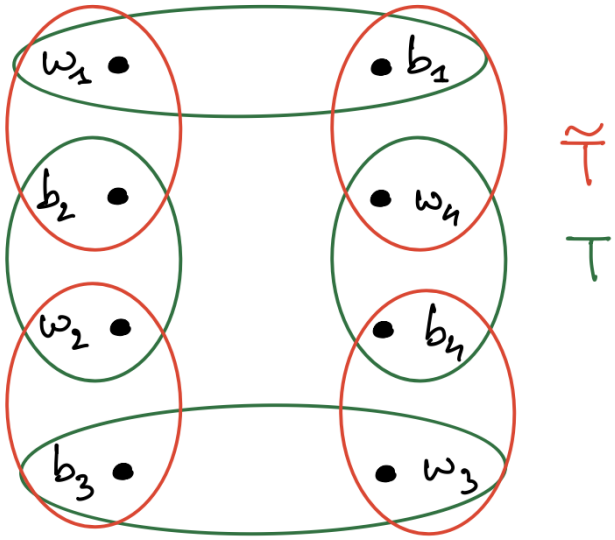
Lemma 7.0.8. *The number of domino tilings of G is equal to $|\det(K)|$.*

Proof. (Part 1 of Lemma [7.0.8]) It is enough to prove that for any two σ,$\widetilde{\sigma}$ corresponding to dimer tilings,

$$\mathrm{sign}(\sigma)k_{1\sigma(1)}...k_{N\sigma(N)}=\mathrm{sign}(\widetilde{\sigma})k_{1\widetilde{\sigma}(1)}...k_{N\widetilde{\sigma}(N)},\tag{7.3}$$

since each non-zero term has modulus 1 and corresponds to a dimer tiling, and where $\mathrm{sign}(\sigma)$ is the signature of the permutation σ.

Let us superimpose the corresponding dimer tilings: $T, \widetilde{T} \subset E$, and consider $T \Delta \widetilde{T} = (T \backslash \widetilde{T}) \cup (\widetilde{T} \backslash T)$ is the symmetric difference of T and $\widetilde{T}$. We set a collection of cycles. We can go from T to $\widetilde{T}$ by a sequence of sliding moves along each of the cycles: replace all the edges of the tiling along that cycle by the other edges along that cycle. Then if for each such moves we keep $\text{sign}(\sigma) k_{1\sigma(1)} ... k_{N\sigma(N)}$ invariant, we have the lemma. Hence we can assume that T and $\widetilde{T}$ differ by <u>one</u> cycle.
For a cycle, such that $b_1 \sim w_l$, given by $b_1 w_1 b_2 w_2 ... b_l w_l$ we can consider the tilings $T = (b_1, w_1)...(b_l, w_l)$ and, $\widetilde{T} = (b_1, w_l)(b_2, w_1)...(b_l, w_{l-1})$. Now, to obtain σ, we compose $\sigma = (1, 2, 3, ..., l) \circ \widetilde{\sigma}$.



Hence $\text{sign}(\sigma) = (-1)^{l+1} \in \text{sign}(\widetilde{\sigma})$. Now we need to prove that, for a simple cycle $b_1 w_1 ... b_p w_p$ enclosing q vertices, we have:

$$\frac{k_{11} ... k_{pp}}{k_{1p} k_{2p} ... k_{p(p-1)}} = (-1)^{q+1+p} \qquad \text{(holds up to relabeling).} \tag{7.4}$$

With (7.4) we will get (7.3) as in this case q is even since the vertices inside admit a perfect matching. To prove (7.4) proceed by an inductive argument, growing the cycle by a sequence of moves (remote cousin of $\oint f = 2i \int \int \overline{\partial} f$).

□

Definition 7.0.9. *We define the* ***full Kasteleyn matrix*** $\widetilde{K}$ *as:*

$$\widetilde{k}_{vw} := \begin{cases} 1, & \text{if } (v,w) \text{ is a horizontal edge.} \\ i, & \text{if } (v,w) \text{ is a vertical edge.} \\ 0, & \text{otherwise.} \end{cases} \tag{7.5}$$

And since vertices can be indexed by (x, y), we have a matrix mapping of functions $f : \{1, ..., m\} \times \{1, ..., n\} \longrightarrow \mathbb{C}$ to other such functions.

Lemma 7.0.10.

$$|\det(\widetilde{K})| = |\det(K)|^2. \tag{7.6}$$

Proof. By reordering vertices,

$$\widetilde{K} = \begin{pmatrix} 0 & K \\ K^T & 0 \end{pmatrix} \tag{7.7}$$

where this changes at most the sign of the determinant. Therefore, $|\det(\widetilde{K})| = |\det(K)|^2$. □

Proof. (Part 2 of Lemma [7.0.8]) Now, let us find an eigenbasis of mn eigenvectors. Using that $\widetilde{K}$ maps {functions $\{1,...,m\} \times \{1,...,n\} \longrightarrow \mathbb{C}$} to itself, we can write:

$$(\widetilde{K}f)(x,y) = f(x+1,y) + f(x-1,y) + if(x,y+1) + if(x,y-1), \tag{7.8}$$

with the conventions that $f(0,\cdot) = f(m+1,\cdot) = 0$ and $f(\cdot,0) = f(\cdot,n+1) = 0$. We guess that eigenfunctions should be of the form $(z^x - z^{-x})(w^y - w^{-y})$ with appropriate values of z,w, i.e. when z is an $(m+1)$-root of 1. This leads to $z = e^{\frac{\pi ij}{m+1}}$ and $w = e^{\frac{\pi ik}{n+1}}$, we then see that the eigenvalues in this case are:

$$(z+z^{-1}) + i(w+w^{-1}) = 2\cos\left(\frac{\pi j}{m+1}\right) + 2i\cos\left(\frac{\pi k}{n+1}\right). \tag{7.9}$$

Finally, we get Kasteleyn's Theorem (7.0.1) using the fact that $\det(K) = \sqrt{\det(\widetilde{K})}$ is a product of the eigenvalues (the eigenvectors are independent, as they have distinct eigenvalues).

□